

The Higgs branch of a compact Calabi–Yau transition at finite Planck mass

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ABSTRACT: When a compact Calabi–Yau threefold degenerates, M-theory on it acquires light charged hypermultiplets whose condensation opens a Higgs branch, locally a hyperkähler quotient without gravity. At finite Planck mass the hypermultiplet moduli space is quaternionic Kähler, its metric is unknown, and its moment maps have no additive constant, hence no Fayet–Iliopoulos parameter. We prove a local normal form for quaternionic Kähler quotients near a fixed point at which the moment map vanishes, under a freeness condition on the normal representation: the quotient is locally the product of the fixed set with the flat hyperkähler quotient of that representation, with relative metric corrections of order r^2 at distance r . Consequently, if exactly k disjoint $(-1, -1)$ -curves have zero area on a face of the nef cone with finite gauge couplings, and their classes satisfy one primitive relation $\sum_i a_i [C_i] = 0$ with all $a_i \neq 0$, then, under a regularity hypothesis on the effective supergravity, the Higgs branch opened by the k light hypermultiplets is locally a smooth factor times $\mathbb{C}^2/\mathbb{Z}_m$, $m = \sum_i |a_i|$, as without gravity. Transverse to the smooth factor, the order- r^2 correction is the curvature of the hypermultiplet manifold restricted to a quaternionic line. A transition with two nodes and a cone over the degree-seven del Pezzo surface has such a face with $k = 4$, giving $\mathbb{C}^2/\mathbb{Z}_4$; where the del Pezzo surface collapses, the Higgs branch defined as the metric completion of the vacua on the smoothed threefold is again of this form. In type IIA string theory, under corresponding hypotheses, the $\mathbb{C}^2/\mathbb{Z}_k$ point that Greene, Morrison and Vafa expected near k homologous vanishing three-cycles persists at finite Planck mass for these smoothings. A census of toric hypersurfaces finds 5605 faces whose light spectrum has one relation, 372 with $k \geq 3$, all with $m = k$.

KEYWORDS: M-Theory, Superstring Vacua, Supergravity Models, Differential and Algebraic Geometry, Supersymmetric Gauge Theory

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1 Introduction

When a compact Calabi–Yau threefold develops singularities, M-theory on it acquires light states, and the vacuum moduli space of the five-dimensional supergravity acquires a new branch along which these states condense. With gravity decoupled, the local structure of such a Higgs branch is a hyperkähler quotient computed from the charges of the light states. In a compactification on a compact threefold the Planck mass is finite, the hypermultiplet scalars take values in a quaternionic Kähler manifold rather than a hyperkähler one [1], and the metric of that manifold is not known [2]. This paper asks what the local space of supersymmetric vacua is at finite Planck mass, and shows that for a class of transitions it can be determined without knowing the metric.

We use the following terms. A *root* of the Higgs branch is a point at which it meets the Coulomb branch, that is, where the charged hypermultiplets become massless. If near a root the Higgs branch is locally a product (vector moduli) $\times$ (neutral hypermultiplets) $\times S$ with S a cone, we call the germ of S at its vertex the *transverse type* of the root. The *rigid model* of a root is the flat hyperkähler quotient of the light hypermultiplets by the gauge group that acts on them, with gravity and all other fields ignored. In the language of stratified spaces the transverse type is the transverse slice to the stratum through the root. Finally, a *Wilsonian* effective action is one for the modes below a fixed mass scale that keeps the light charged states as fields, as opposed to the moduli-space action obtained by integrating them out.

The example. The threefold X_9 of [3, 4], recalled in Section 2.1, has two ordinary double points and one cone over the degree-seven del Pezzo surface $S_7 = \text{Bl}_2 \mathbb{P}^2$; its crepant resolution $\widehat{X}_9$ has Hodge numbers $(h^{1,1}, h^{2,1}) = (6, 122)$ and its smoothing $X_{9,t}$ has $(3, 123)$. Physically, the two nodes contribute two charged hypermultiplets and the del Pezzo cone the rank-one E_2 theory. This is the strongly coupled five-dimensional superconformal theory, without a Lagrangian description, obtained from M-theory on a local del Pezzo surface of degree seven; in the same series E_1 corresponds to $\mathbb{P}^1 \times \mathbb{P}^1$ and E_0 to $\mathbb{P}^2$. Write C_1, C_2 for the exceptional curves of small resolutions of the two nodes, u_1, u_2 for their smoothing parameters ($xy - zw = u_i$) and s for the deformation parameter of the del Pezzo cone (Section 2.1). The three sectors are charged under the same two bulk vector multiplets, and every global first-order deformation of X_9 satisfies $u_1 + s = u_2 + s = 0$, so none of them can be Higgsed alone: every first-order smoothing of X_9 smooths all three singular points at once. The rigid model of this transition has transverse type

$$\mathbb{C}^2/\mathbb{Z}_4, \quad \mathbb{C}[X, Y, s]/(XY - s^4), \quad (1.1)$$

an A_3 surface singularity [4, eq. (3.10)]. The coupling of the three sectors is a gravitational effect. The inverse gauge couplings of the vector multiplets are the Hodge norms of the corresponding two-forms. Along any path on which the volume of the threefold grows while the areas of C_1, C_2 and of all curves in the del Pezzo surface stay bounded, some combination of the classes that couple the sectors acquires a divergent norm and its canonically normalised

coupling to the local fields vanishes. The joint constraint is then never retained; at most $u_1 = u_2$ survives [4, Sect. 4, Thm. 4.3]. The question is therefore only meaningful at finite Planck mass.

Results. First, we prove a local normal form for quaternionic Kähler quotients (Theorem 4.5). Let a compact connected group G act isometrically on a quaternionic Kähler manifold M of dimension at least eight and nonzero scalar curvature, which need not be complete, and let p lie on a component N of the fixed set on which the moment map vanishes. If G acts locally freely on the nonzero part of the zero set of the flat hyperkähler moment map μ_0 of the normal isotropy representation V , then near $[p]$ the quotient is homeomorphic, compatibly with its stratification, to $N \times V///G$, where $V///G = \mu_0^{-1}(0)/G$ is the flat hyperkähler quotient. If the action there is free, the metric tangent cone is the flat product, and the metric agrees with it up to relative corrections of order r^2 at distance r from $[p]$ (Proposition 4.7); the rate is attained. The statement holds for either sign of the scalar curvature, so it applies to positive quaternionic Kähler manifolds as well. Behind this lies an elementary observation (Lemma 4.1 and Remark 4.2): the moment map of a quaternionic Kähler manifold has no constants of integration, and at a fixed point it is determined by the isotropy representation. A hyperkähler quotient can be resolved or deformed by moving the level of the moment map, a Fayet–Iliopoulos parameter; at finite Planck mass no such parameter exists, and a nonzero value of the moment map at a fixed point is the same thing as a gauging of (a subgroup of) the $SU(2)$ R-symmetry there. The normal form and the tangent-cone estimate apply, at the level $\mu(p)$, to hyperkähler and Kähler quotients at a fixed point (Remark 4.9).

When the flat quotient $V///G$ is four-dimensional and the action is free off the origin, we also identify the correction of order r^2 transverse to N . Suppose that V contains a quaternionic line that lies in $\mu_0^{-1}(0)$ and is orthogonal to the G -orbits through its points, as it does for the torus actions on $\mathbb{H}^k$ with one relation among the weights considered below. That line maps onto the flat quotient and gives an orbifold chart of the transverse slice. The metric in this chart is not known to be twice differentiable at the vertex, but it has an expansion to second order, and we call the curvature at the vertex of its second-order truncation the *leading curvature*. It is the curvature of M at p restricted to that line (Proposition 4.12): the part of the curvature fixed by supergravity contributes constant curvature 4ν , where ν is the reduced scalar curvature of M (with $\nu = 1$ for $\mathbb{H}\mathbb{P}^n$ of sectional curvature in $[1, 4]$), and the hyperkähler part of the curvature contributes an anti-self-dual Weyl tensor, with respect to the orientation in which the Kähler forms ω_a are self-dual (self-dual in the opposite orientation customary in four-dimensional quaternionic Kähler geometry).

Second, the physical statement holds for every face with one relation (Theorem 5.2). Suppose that on an open face F of the nef cone of a compact Calabi–Yau threefold $\hat{X}$ exactly k disjoint $(-1, -1)$ -curves have zero area, and that their classes satisfy a single primitive relation $\sum_i a_i [C_i] = 0$ with every $a_i \neq 0$, that F is open in the annihilator of their classes, and that the gauge kinetic matrix of the vector multiplets is positive definite on F (Section 3.1). Assume the regularity hypothesis of Section 3.4, whose main content is that there is a smooth

Wilsonian hypermultiplet manifold containing the k light states and that the Coulomb branch through the root consists of supersymmetric vacua. Then the Higgs branch at the root has transverse type $\mathbb{C}^2/\mathbb{Z}_m$ with $m = \sum_i |a_i|$, the same as its rigid model, and the metric of its hypermultiplet factor is the flat orbifold metric of $\mathbb{C}^2/\mathbb{Z}_m$ times the flat metric on the tangent space of N , up to relative corrections of order r^2 , where r is the distance from the root. For k nodes whose vanishing three-spheres are homologous up to sign, $m = k$.

Third, X_9 has such a face (Section 2). On a three-dimensional open face F of the nef cone of $\widehat{X}_9$ exactly four irreducible curves have zero area, the two nodal curves and the two exceptional curves of the del Pezzo surface, and they are disjoint $(-1, -1)$ -curves (Proposition 2.1); so the massless spectrum, in all degrees and genera, consists of four hypermultiplets. No divisor collapses and all gauge couplings stay finite on F , and the resulting threefold with four nodes smooths, in the direction allowed by its single relation, to a smoothing of X_9 deformation equivalent to $X_{9,t}$ (Proposition 2.2). The four vanishing charges and their relation were found from the mirror periods of a four-node degeneration in [4, Sects. 5.1–5.2, Prop. 5.1], and the corresponding mirror family was related there to that of $X_{9,t}$ [4, Prop. 5.2]. What is new here is the statement in five-dimensional M-theory, that on F exactly these curves and no others have zero area, together with a direct proof that the four-node threefold smooths to $X_{9,t}$. At a root over F the light states are four free hypermultiplets of a rank-three torus with relation $a = (1, 1, -1, 1)$, and the local space of supersymmetric Minkowski vacua is (Theorem 5.3)

$$\mathcal{V}_x \cong (\mathcal{K}_\phi \times N) \cup (F_\phi \times N \times \mathbb{C}^2/\mathbb{Z}_4), \quad (1.2)$$

where $\mathcal{K}_\phi$ is the germ of the Coulomb branch, F_ϕ the germ of the face and N the manifold of 123 neutral hypermultiplets, the two branches being glued along $F_\phi \times N \times \{0\}$. The transverse type is the rigid one (1.1): gravity does not remove the A_3 point or change its type. The leading curvature of the transverse $\mathbb{C}^2/\mathbb{Z}_4$ is the constant curvature 4ν fixed by supergravity plus a $\mathbb{Z}_4$ -invariant anti-self-dual Weyl tensor; the latter is the only compactification-dependent datum at this order. It lies in a space of dimension three, every element of which comes from some invariant algebraic curvature tensor of hyperkähler type (Proposition 5.5); whether each comes from a quaternionic Kähler manifold with this torus action is not addressed. Curvature tensors of hyperkähler type correspond to quartic polynomials. One of the three directions, invariant under a triholomorphic circle, is realised by the curvature of the Taub–NUT space at its nut; the other two come from the generators X, Y of the ring (1.1), which are quartic because $m = 4$.

A census of the hypersurfaces from favourable Kreuzer–Skarke polytopes with $h^{1,1} \leq 5$ and from the favourable members of a random sample with $h^{1,1} = 6$ (Section 7.4) finds 5605 faces of toric Kähler cones whose light spectrum, computed up to degree six, has a single relation: 5233 flop walls with $k = 2$, and 372 faces with $k = 3, 4$ or 5 in 210 polytopes. In none of them does a coefficient $|a_i| \geq 2$ occur, so $m = k$ throughout; whether faces with $m \neq k$ exist at all remains open. For 4092 of the 5605 faces toric data show that the contraction has exactly k nodes, the light states being disjoint $(-1, -1)$ -curves; in the others they do not

decide it.

The del Pezzo point, where the whole surface collapses and the E_2 theory appears, lies on the facet of the closure of F on which the del Pezzo surface has zero volume. Call the part of the Higgs branch on which M-theory is compactified on the smooth threefold $X_{9,t}$ the *regular Higgs branch*. The facet is not a boundary of it: the regular Higgs branch continues across the facet, and its vacua immediately beyond the facet end at roots where a surface of a flopped threefold collapses and the E_1 theory, together with the two nodes, appears (Section 7.2). Since the E_2 theory has no Lagrangian, we define the Higgs branch at the del Pezzo point as the metric completion of the regular Higgs branch. Because the hypermultiplet metric of that part does not depend on the vector moduli (Lemma 3.3), the germ computed over F is carried across the facet, and the completion again has transverse type $\mathbb{C}^2/\mathbb{Z}_4$ (Theorem 5.6).

Fourth, in type IIA string theory on $\hat{X}$ in four dimensions, under the corresponding four-dimensional regularity hypothesis, at large volume, the Wilsonian vacuum space at the locus where the D2-branes on the curves C_i become massless has the same structure: the hypermultiplet factor of the Higgs branch is locally $N \times \mathbb{C}^2/\mathbb{Z}_m$, quaternionic Kähler off $N \times \{0\}$ and flat up to relative corrections of order r^2 (Theorem 6.2). The massless states carry no D0-charge, the four-dimensional vacuum conditions [5] replace those of five-dimensional supergravity, and at large volume the special Kähler metric is the five-dimensional gauge kinetic matrix divided by 4 vol (Proposition 6.3), so the positivity assumed in five dimensions carries over. The k vanishing three-spheres of the smoothing satisfy $[S_i] = a_i[S]$ (Lemma 6.4); for unit coefficients they are homologous up to sign. If in addition the Higgs vacua are type IIA compactifications on the smoothing, with hypermultiplet moduli space bi-Lipschitz equivalent to the quotient with its quotient metric, then the type IIA hypermultiplet moduli space at finite Planck mass keeps its A_{k-1} point near the k homologous vanishing three-cycles and is quaternionic Kähler around it (Corollary 6.5): this is the statement Greene, Morrison and Vafa expected, for the smoothings of the contractions considered here.

Outline of the proofs. The normal form rests on three facts about a fixed point p with vanishing moment map. At a fixed point the moment map is the $\mathfrak{sp}(1)$ -part of the isotropy (Lemma 4.1), so $\mu(p) = 0$ forces the isotropy to preserve a flat hyperkähler structure on $T_p M$. Along normal geodesics the moment map is the flat one up to terms of order four, with no cubic term (Lemma 4.4), because the fundamental vector field is a Jacobi field vanishing at the starting point. Blowing up along N , the rescaled zero set is transverse on the link at $r = 0$, and the flow of a connection carries the flat cone to the true zero set; the absence of the cubic term is what makes the deviation of order r^2 rather than r . The general physical theorem then reduces to its flat model: for a single relation with nonzero coefficients the torus acts freely off the origin of the zero set of the flat moment map, and the flat quotient is $\mathbb{C}^2/\mathbb{Z}_m$ (Lemma 5.1). At the root the moment map vanishes because the Coulomb branch through it consists of supersymmetric vacua (the first clause of the hypothesis) and the gauge kinetic matrix is nondegenerate there, so that the vacuum conditions force every moment map to vanish. For X_9 the face is found by a contraction argument: every $J \in F$ is a class

pulled back from an ample class on X_9 plus a positive multiple of a nef class that restricts to the hyperplane class on the del Pezzo surface, so a curve of zero area is contracted by the resolution and, if it lies in the surface, is one of its exceptional curves.

Relation to earlier work. With gravity decoupled, the $\mathbb{C}^2/\mathbb{Z}_k$ structure of the hypermultiplet moduli space near k homologous vanishing three-cycles was derived from D-instanton sums by Ooguri and Vafa [6] and by Seiberg and Shenker [7], the multiplicity was identified with a Gopakumar–Vafa invariant [8], and smooth hyperkähler model geometries of this kind were constructed rigorously for $k = 1$ in [9], where the extension to multiplicity k is indicated. Greene, Morrison and Vafa obtained $\mathbb{C}^2/\mathbb{Z}_k$ for k hypermultiplets with one relation of unit coefficients [10, Sect. 3, eqs. (3.13)–(3.18)] and expected that with gravity its qualitative features persist while the metric is corrected [10, Sect. 6], a statement about the four-dimensional hypermultiplet space. Theorem 5.2 and Proposition 4.7 make the five-dimensional counterpart precise, under the regularity hypothesis, for every face with one relation with nonzero coefficients, and Corollary 6.5 the four-dimensional statement itself, for the smoothings considered here and under the identification (W-IIA) of Section 6.4. Mohaupt and Saueressig constructed five-dimensional gauged supergravities for M-theory conifold transitions with a model hypermultiplet metric [2]; the germ found here does not depend on such a choice. On the mathematical side, quaternionic Kähler reduction is due to Galicki and Lawson [1] and the correspondence with hyperkähler cones to Swann [11]. Dancer and Swann described a singular quaternionic Kähler quotient as a union of smooth pieces, some quaternionic Kähler and some locally Kähler, and as a union of quaternionic Kähler orbifolds [12, Thm. 4.6, Props. 4.4 and 5.1, Thm. 6.1], and showed that for compact connected M and G each component is a complete length space [12, Thm. 7.1]; through a fixed point with vanishing moment map the piece is the fixed component N (Lemmas 4.1 and 4.3). These works describe the pieces but not how they fit together: the frontier condition is not established [12, Sect. 2], and no local model near a singular piece is given. In symplectic geometry that role is played by the local normal form of Marle and of Guillemin and Sternberg [13, 14], on which the stratification of singular symplectic quotients by Sjamaar and Lerman [15] rests. The hyperkähler analogue of that local model is not a normal form, since a hyperkähler equivalence is an isometry [16, p. 149]; see also [17, Sect. 1]. For hyperkähler quotients of positive-definite hyperkähler manifolds on which the complexified group acts, that is, on which the vector fields $I_1 X_\xi$ are complete, Mayrand gives a complex-analytic local model at every point of the quotient [17, Thm. 1.4], and the product form of the germ along a fixed component was conjectured for semi-free circle actions by Fang [18, Rem. 3.2]. For negative scalar curvature the Swann bundle is pseudo-hyperkähler, so the model of [17] does not apply, and Theorem 4.5 is proved directly on the quaternionic Kähler manifold; for positive scalar curvature that model describes the complex-analytic germ of the hyperkähler cone, which is not the metric statement along N proved here. Theorem 4.5 is accordingly a model up to homeomorphism of germs, with metric estimates, and Proposition 4.7 describes the tangent cone of the orbit length metric at such points without assuming compactness; for compact connected M and G this metric is that of

Dancer and Swann.

Scope. The results of Section 4, among them Theorem 4.5 and Proposition 4.12, are theorems about quaternionic Kähler quotients and are proved unconditionally, as are Propositions 2.1, 2.2 and 6.3 and Lemmas 5.1 and 6.4. Lemmas 3.3 and 5.4 and Theorems 5.2, 5.3 and 5.6 are proved under Hypothesis 3.2; Theorem 6.2 and Corollary 6.5 under Hypothesis 6.1 and, for the corollary, the identification (W-IIA) of Section 6.4. Proposition 5.5 is proved for the manifold M of Hypothesis 3.2; Appendix A describes an independent computer check of it and of Proposition 4.12. The census of Section 7.4 rests on computer calculations of genus-zero invariants. Theorem 5.6 describes the metric completion of the part of the Higgs branch over the smooth threefold $X_{9,t}$, over a region of the Kähler cone of $X_{9,t}$ that contains the del Pezzo facet, not the vacuum space of the interacting theory at the del Pezzo point. The statements about the other roots of X_9 in Section 7 concern their light spectra and rigid models only. All statements about the vacuum space are topological or metric; no complex-analytic structure is claimed.

Organisation. Section 2 recalls X_9 and proves the statements about the face. Section 3 states the setting for a general face with one relation, the vacuum conditions and the regularity hypothesis. Section 4 proves the normal form and the tangent-cone statement and identifies the leading curvature. Section 5 proves the theorems on the Higgs branch and computes the leading correction for X_9 . Section 6 treats type IIA string theory in four dimensions. Section 7 compares with the rigid picture, describes the other roots of X_9 , compares with the instanton picture, reports the census of one-relation faces, discusses other effects at finite Planck mass, and lists open problems. Appendix A prints the intersection data, the kinetic determinant, the Gopakumar–Vafa data and the charge conventions, and describes an independent check of the leading correction.

2 The transition and a face with four contracted curves

2.1 The threefold X_9 and its transition

We recall the facts about X_9 used below; all of them are established in [3, 4].

The resolution $\widehat{X}_9$ is a smooth anticanonical hypersurface in a toric fourfold whose fan is built on the polytope with vertices $v_0, \dots, v_8$ of [4, eq. (3.1)]. For a vertex v_i we write D_i for the restriction to $\widehat{X}_9$ of the corresponding toric divisor. Blowing down the exceptional locus gives a morphism $\pi : \widehat{X}_9 \rightarrow X_9$ onto a Calabi–Yau threefold X_9 with three singular points: two ordinary double points and one cone over the degree-seven del Pezzo surface $S_7 = \text{Bl}_2 \mathbb{P}^2$. The exceptional locus of π consists of the two nodal curves C_1, C_2 and a surface $E \cong S_7$, the divisor of an interior lattice point of a two-dimensional face of the polytope. We write e_1, e_2 for the two exceptional curves of E and h for the pullback of the hyperplane class of $\mathbb{P}^2$, so that $h - e_1 - e_2$ is the third (-1) -curve of E .

The singular threefold X_9 has an explicit smoothing $X_{9,t}$ [4, Sect. 3.1]. Near the three singular points, write the nodes as $xy - zw = u_i$, $i = 1, 2$, and let s be the coordinate on

the reduced local deformation space of the del Pezzo cone of [4, eqs. (2.2), (2.3)], denoted β there; in the E_2 theory it is the Cartan component of the moment map of its $SU(2)$ flavour symmetry. The image of every global first-order deformation of X_9 in these local parameters lies on the line

$$u_1 + s = u_2 + s = 0, \quad (2.1)$$

and the smoothing moves along it in the direction $(u_1, u_2, s) = t(-1, -1, 1)$, smoothing all three points at once [4, eqs. (3.4), (3.9)]; the statement about all global first-order deformations follows from the proof of [4, Thm. 2.2], which applies to the full space of local first-order deformations. The reason is that the two nodal curves and the class $e_1 - e_2$ of the del Pezzo surface satisfy $\iota_*(e_1 - e_2) = C_1 + C_2$ in $H_2(\widehat{X}_9; \mathbb{Z})$, so the three local sectors are charged under the same two vector multiplets. The transition changes the Hodge numbers from $(h^{1,1}, h^{2,1}) = (6, 122)$ to $(3, 123)$ [4, eqs. (1.2), (3.5)]: three vector multiplets are lost, two Higgsed by the nodal and del Pezzo hypermultiplets and one because the Coulomb direction of the E_2 theory disappears, and one neutral hypermultiplet appears.

2.2 A face on which four curves contract

Let $H_1, \dots, H_6$ be the integral basis of $H^2(\widehat{X}_9; \mathbb{Z})$ formed by the generators of the ambient nef cone restricted to $\widehat{X}_9$ [4, eq. (C.1)], and $\beta_1, \dots, \beta_6$ the dual curve classes. Writing $\iota : E \rightarrow \widehat{X}_9$ for the inclusion, in this basis

$$C_2 = \beta_1, \quad C_1 = \beta_5, \quad \iota_* e_2 = \beta_6, \quad \iota_* e_1 = \beta_1 + \beta_5 + \beta_6, \quad \iota_*(h - e_1 - e_2) = \beta_2, \quad (2.2)$$

and $H_3 = D_1$, $H_4 = D_0$ are the two classes pulled back from X_9 ; the divisors D_0, D_1 do not meet the exceptional locus. For J in the Kähler cone the gauge kinetic matrix of the vector multiplets is proportional to the Hodge metric $\int \omega_I \wedge * \omega_K$ on harmonic two-forms. With $L(J) = (\kappa_{IKM} J^M)_{I,K=1}^6$ the full 6×6 matrix on $H^2(\widehat{X}_9)$, $v = LJ$ and $\text{vol} = J^3/6$, it is $\mathcal{G} = -L + vv^\top / (4 \text{vol})$ [4, eqs. (4.1), (4.6)]. For curve classes $\gamma_1, \dots, \gamma_k$ we write $\text{ann}(\gamma_1, \dots, \gamma_k)$ for the subspace of $H^2(\widehat{X}_9; \mathbb{R})$ on which they all vanish. Set

$$F := \{ J = y_2 H_2 + y_3 H_3 + y_4 H_4 : y_2, y_3, y_4 > 0 \}. \quad (2.3)$$

Since $H_2 \cdot e_i = 0$ and $H_2 \cdot (h - e_1 - e_2) = 1$, one has $H_2|_E = h$, and hence $J|_E = y_2 h$ for $J \in F$.

Proposition 2.1. $\overline{F} = \text{Nef}(\widehat{X}_9) \cap \text{ann}(C_1, C_2, e_1, e_2)$, and for $J \in F$:

1. an irreducible curve $C \subset \widehat{X}_9$ has $J \cdot C = 0$ if and only if $C \in \{C_1, C_2, e_1, e_2\}$; these four curves are pairwise disjoint and each has normal bundle $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$;
2. every prime divisor has positive volume; in particular $\text{vol}(E) = \frac{1}{2} J^2 \cdot E = y_2^2/2$;
3. the gauge kinetic matrix $\mathcal{G}$ is positive definite.

Proof. Since H_2, H_3, H_4 are nef, $J \cdot C = 0$ forces $H_2 \cdot C = D_1 \cdot C = D_0 \cdot C = 0$. The classes $y_3 D_1 + y_4 D_0$ with $y_3, y_4 > 0$ are exactly $\pi^* \text{Amp}(X_9)$. Indeed $\pi^* \text{Nef}(X_9) = \text{Nef}(\widehat{X}_9) \cap \pi^* H^2(X_9; \mathbb{R}) = \text{cone}(D_0, D_1)$ [4, Prop. 4.1], and every line bundle L on $\widehat{X}_9$ of degree zero on all curves in the fibres of π is pulled back from X_9 : since $K_{\widehat{X}_9} = 0$ and every line bundle is π -big for birational π , the relative basepoint-free theorem [19, Thm. 3.24], with $\Delta = 0$ and $a = 1$, shows that L^b and L^{b+1} are π -globally generated for $b \gg 0$; being of degree zero on the connected fibres, they are trivial there and descend, as $\pi_* \mathcal{O}_{\widehat{X}_9} = \mathcal{O}_{X_9}$ for the normal variety X_9 , and hence so does $L = L^{b+1} \otimes L^{-b}$, as in [19, proof of Thm. 3.7, Step 9]. By Kleiman's criterion [20] the descended class of $y_3 D_1 + y_4 D_0$ is ample for $y_3, y_4 > 0$. Hence C is contracted by π , that is $C \subset C_1 \cup C_2 \cup E$. On E one has $J|_E = y_2 h$, and the irreducible curves of E with $h \cdot C = 0$ are e_1 and e_2 . The two nodes are distinct points of X_9 away from the image of E , and $e_1 \cdot e_2 = 0$ on E , so the four curves are disjoint. For e_i one has $N_{e_i/E} = \mathcal{O}(-1)$ and $E \cdot e_i = K_E \cdot e_i = -1$, and the extension of normal bundles splits; the C_i are the exceptional curves of small resolutions of ordinary double points. With (2.2) this proves (1) and $\overline{F} \subseteq \text{Nef}(\widehat{X}_9) \cap \text{ann}(C_1, C_2, e_1, e_2)$; the reverse inclusion holds because β_2, β_3 and $\beta_4 + \beta_5 + \beta_6$ are effective (their genus-zero Gopakumar–Vafa invariants are 1, -2 and 10), which forces $y_2, y_3, y_4 \geq 0$.

For (2), a class $J \in F$ with rational coordinates is nef and big, so by the basepoint-free theorem [19, Thm. 3.3] some multiple defines a birational morphism $\pi_F : \widehat{X}_9 \rightarrow X_F$ with $J = \pi_F^* A$, A ample. If a prime divisor D had $J^2 \cdot D = A^2 \cdot \pi_{F*} D = 0$, then $\dim \pi_F(D) \leq 1$ and D would lie in the exceptional locus of π_F , which by (1) is the union of four curves. At an irrational point write $J = \sum_i a_i J_i$ with $a_i > 0$ and rational $J_i \in F$; then $J^2 \cdot D = \sum_{i,j} a_i a_j J_i \cdot J_j \cdot D \geq \sum_i a_i^2 J_i^2 \cdot D > 0$, because the mixed terms are nonnegative for nef J_i, J_j and effective D . The value of $\text{vol}(E)$ follows from $J|_E = y_2 h$.

For (3), since $v = LJ$ we have $v^\top L^{-1} v = J^\top LJ = 6 \text{vol}$, and the matrix determinant lemma gives $\det \mathcal{G} = \det(-L)(1 - v^\top L^{-1} v / (4 \text{vol})) = -\frac{1}{2} \det L$, where $\det(-L) = \det L$ because the size is even. In the coordinates (y_2, y_3, y_4) ,

$$\det L(J) = -y_2(2y_3 + 3y_4)(y_2 + 2y_4) P(y_2, y_3, y_4), \quad (2.4)$$

with P the polynomial with positive coefficients printed in (A.3). So $\det \mathcal{G} > 0$ on F . On $\overline{F} \setminus \mathbb{R}_{\geq 0} H_2$ the volume is positive and $\mathcal{G}$ is a limit of Hodge norms from the Kähler cone, hence positive semidefinite, and it is positive definite wherever its determinant is nonzero. $\square$

Four flop walls meet along F , one for each of the curves in (1). When the four curves are used as the curves $C_1, \dots, C_4$ of Section 3.1 we put $C_3 = e_1$ and $C_4 = e_2$.

2.3 The contraction and its smoothing

The contraction $\pi_F : \widehat{X}_9 \rightarrow X_F$ of the proof of Proposition 2.1 has four ordinary double points, the images of C_1, C_2, e_1, e_2 , and the classes of these curves satisfy the single relation

$$[C_1] + [C_2] - [e_1] + [e_2] = 0, \quad (2.5)$$

all of whose coefficients have absolute value one. By Friedman's criterion [21] and the unobstructedness of its deformations [22], X_F is smoothable, with a one-dimensional space of local smoothing directions compatible with (2.5).

Write u_{C_i}, u_{e_i} for the local smoothing parameters of the four double points of X_F (for a node $xy - zw = u$), and (u_1, u_2, s) for the local parameters of X_9 of Section 2.1.

Proposition 2.2. *A smoothing of X_F in the direction permitted by (2.5) has general fibre a smoothing of X_9 in the direction $(u_1, u_2, s) = t(-1, -1, 1)$ of Section 2.1; in particular it is deformation equivalent to $X_{9,t}$.*

Proof. The morphism $X_F \rightarrow X_9$ contracts the image $\overline{E} \cong \mathbb{P}^2$ of E to a rational singularity, so deformations of X_F blow down to deformations of X_9 [23, Prop. 11.4]. The permitted direction is $(u_{C_1}, u_{C_2}, u_{e_1}, u_{e_2}) = t(-1, -1, 1, -1)$. The two nodes of X_9 are the same germs as the nodes C_1, C_2 of X_F , so the induced deformation has $u_i = u_{C_i} = -t \neq 0$. The local image of every global first-order deformation of X_9 lies on the line (2.1), by the proof of [4, Thm. 2.2], so $s = t + O(t^2) \neq 0$ for small $t \neq 0$ as well. Hence the induced deformation smooths both nodes and the del Pezzo cone point, and its general fibre $X_{9,t'}$ is smooth. The induced morphism $f : X_{F,t} \rightarrow X_{9,t'}$ between smooth Calabi–Yau threefolds satisfies $K_{X_{F,t}} = f^* K_{X_{9,t'}}$, so it is crepant; a crepant exceptional divisor over a smooth target is impossible, so f is small; and a small birational morphism onto a smooth, hence $\mathbb{Q}$ -factorial, variety is an isomorphism. $\square$

By Proposition 2.1(1), and the vanishing of multiple-cover and higher-genus contributions of isolated $(-1, -1)$ -curves (Appendix A), the charged massless spectrum at a point of F consists of four hypermultiplets, one for each curve, whose charge matrix has rank three and Smith invariants $(1, 1, 1)$. Condensing them Higgses three vector multiplets and leaves one neutral hypermultiplet, which matches $(h^{1,1}, h^{2,1}) = (6, 122) \rightarrow (3, 123)$.

3 Supersymmetric vacua at a face with one relation

3.1 The setting

Let $\widehat{X}$ be a smooth projective Calabi–Yau threefold and F an open face of its nef cone, that is, the relative interior of a face. We assume:

- (S1) for $J \in F$ one has $J^3 > 0$, the irreducible curves C with $J \cdot C = 0$ are exactly k pairwise disjoint $(-1, -1)$ -curves $C_1, \dots, C_k$, and F is open in the annihilator $\text{ann}(C_1, \dots, C_k) \subset H^2(\widehat{X}; \mathbb{R})$ of their classes;
- (S2) the classes of the C_i satisfy exactly one relation $\sum_i a_i [C_i] = 0$, with a primitive and every $a_i \neq 0$, and the lattice $\Lambda \subset \mathbb{Z}^k$ of charge vectors $(D \cdot C_1, \dots, D \cdot C_k)$, $D \in H^2(\widehat{X}; \mathbb{Z})$, is saturated, so that $\Lambda = a^\perp \cap \mathbb{Z}^k$ has rank $k - 1$;
- (S3) the gauge kinetic matrix of the vector multiplets, the Hodge metric $\int \omega \wedge *_J \omega'$ on harmonic two-forms, is positive definite on F .

We call a face satisfying (S1)–(S3) a *face with one relation*. On F the Hodge metric is a limit of Hodge metrics on the Kähler cone, hence positive semidefinite, so (S3) is equivalent to its nondegeneracy there. Saturation in (S2) only normalises the gauge torus: in general the torus acts on the k hypermultiplets below through its image, whose cocharacter lattice is $a^\perp \cap \mathbb{Z}^k$, and a finite subgroup acting trivially on them does not change the germs studied here.

Under (S1) a rational class in F is nef and big and defines a contraction $\pi_F : \widehat{X} \rightarrow X_F$ with k ordinary double points, the images of the C_i . Since all a_i are nonzero, X_F has a first-order smoothing by Friedman’s criterion [21], its deformations are unobstructed [22], and its smoothings X_t satisfy $h^{1,1}(X_t) = h^{1,1}(\widehat{X}) - (k-1)$ and $h^{2,1}(X_t) = h^{2,1}(\widehat{X}) + 1$. In M-theory on $\widehat{X}$ with Kähler class $J \in F$, M2-branes on the C_i give k massless hypermultiplets, one for each curve: an isolated $(-1, -1)$ -curve has genus-zero Gopakumar–Vafa invariant 1 and no multiple-cover or higher-genus contributions [24, 25]. They carry charges in Λ under the torus T^{k-1} that is the image of $H^2(\widehat{X}; \mathbb{R})/H^2(\widehat{X}; \mathbb{Z})$ in $U(1)^k$. The k flop walls through F make the cubic prepotential of $\widehat{X}$ only twice continuously differentiable along F . Writing $Z_i = J \cdot C_i$ for the central charges of the light states and keeping them as fields, the prepotential of the Wilsonian theory in the chamber where all $Z_i > 0$ is

$$\mathcal{F}_W = \mathcal{F}_{\widehat{X}} + \frac{1}{12} \sum_i Z_i^3, \quad (3.1)$$

the Chern–Simons levels shifting by $\frac{1}{2}q \otimes q \otimes q$ for each hypermultiplet of charge q [26, 27]. It is a polynomial with the same Hessian as $\mathcal{F}_{\widehat{X}}$ at $Z = 0$.

For X_9 , Propositions 2.1 and 2.2 of Section 2 show that $\widehat{X} = \widehat{X}_9$ and the face (2.3) satisfy (S1)–(S3) with $k = 4$ and $a = (1, 1, -1, 1)$; see the proof of Theorem 5.3.

3.2 Gauged supergravity and its Minkowski vacua

M-theory on $\widehat{X}$ gives five-dimensional $N = 1$ supergravity with $n_V = h^{1,1}(\widehat{X}) - 1$ vector multiplets and $n_H = h^{2,1}(\widehat{X}) + 1$ neutral hypermultiplets; for $\widehat{X}_9$ these are 5 and 123. The vector-multiplet scalars are the very special real coordinates h^I , $I = 1, \dots, n_V + 1$, on the hypersurface $\frac{1}{6}\kappa_{IKM}h^I h^K h^M = 1$ [28]; the overall volume of $\widehat{X}$ sits in the universal hypermultiplet. The hypermultiplet scalars take values in a quaternionic Kähler manifold M of negative scalar curvature. Supergravity fixes only the reduced scalar curvature of M , in units of the five-dimensional Planck mass; the hyperkähler part of the curvature is not fixed. We measure distances on M in Planck units. The transverse correction of order r^2 found below is given by the curvature of M at the root (Proposition 4.12), which may be large where the neutral moduli approach a degeneration. At a point where charged states become massless the effective theory includes them as further hypermultiplets, and the vectors gauge isometries of M with Killing vectors k_I and quaternionic moment maps P_I ; up to normalisation these are the moment maps μ_ξ of Section 4 for ξ the generator dual to the I -th basis divisor.

For constant scalars and vanishing field strengths, the fermion shifts of the gravitino, gaugini and hyperini give the conditions [4, eq. (2.9)]; compare the ground-state conditions of [29, eqs. (5.17)–(5.18)]:

$$h^I P_I = 0, \quad h_x^I P_I = 0, \quad h^I k_I = 0, \quad (3.2)$$

where h_x^I are the tangent vectors to the very special real hypersurface. The gaugino shift is proportional to $g^{xy}h_y^I P_I$, with g_{xy} the vector-multiplet metric; where g_{xy} is nondegenerate its vanishing is equivalent to the second condition. The vectors h^I and h_x^I always span $\mathbb{R}^{n_V+1}$, so where the vector-multiplet metric is nondegenerate the conditions become

$$P_I = 0 \quad \text{for all } I, \quad h^I k_I = 0 \quad (3.3)$$

[4, eq. (2.10)]; the four-dimensional analogue is [5, Sect. 2.4.2]. The potential vanishes on these solutions. The action of M-theory also contains higher-derivative terms, among them the supersymmetric completion of $A \wedge \text{tr } R \wedge R$ [30]. That they do not change the set of such vacua is part of Hypothesis 3.2 below. It is supported by the observation that on a background with constant scalars and vanishing field strengths the auxiliary fields of the Weyl multiplet are set to zero by the supersymmetry variations and by Lorentz invariance, so that the auxiliary equation of the vector multiplets still reduces to the vanishing of the moment maps.

3.3 The local vacuum space

We write N for the quaternionic Kähler manifold of neutral hypermultiplets of $\widehat{X}$, of quaternionic dimension $h^{2,1}(\widehat{X}) + 1$, and call $b \in N$ *smooth* if N is a smooth manifold near b , that is, if b lies over a point of the complex-structure moduli space of $\widehat{X}$ away from the loci where $\widehat{X}$ acquires further singularities.

Definition 3.1. Let $\phi \in F$ be normalised by $J^3 = 6$ and $b \in N$ a smooth point. The *root point* $x = (\phi, b, 0)$ is the vacuum with vector moduli ϕ , neutral hypermultiplets b and vanishing charged hypermultiplets. The *local vacuum space* $\mathcal{V}_x$ is the germ at x of the set of constant-scalar, zero-field-strength solutions of (3.3) in the theory of Hypothesis 3.2 below, modulo gauge equivalence, regarded as a stratified topological space.

The *Coulomb branch* through x is the part of $\mathcal{V}_x$ on which the charged hypermultiplets vanish; we write $\mathcal{K}_\phi$ for its vector-multiplet germ at ϕ . It is the germ at ϕ of the hypersurface of normalised vector-multiplet scalars, divided by the walls $Z_i = 0$ into chambers subject to $\sum_i a_i Z_i = 0$; among these chambers are the Kähler cone of $\widehat{X}$ and those of its flops along the C_i . The *Higgs branch* is the closure of the part of $\mathcal{V}_x$ on which the gauge group acting on the charged hypermultiplets is broken to a finite group, and a *mixed branch* would be a component on which some, but not all, of the charged hypermultiplets are nonzero. We write F_ϕ for the germ at ϕ of $F \cap \{J^3 = 6\}$. On the Higgs branch away from its root the compactification is M-theory on a smoothing X_t , $t \neq 0$: we call this part the *regular Higgs branch* $\mathcal{H}^{\text{reg}}$. At each of its points the vector moduli are normalised Kähler classes of X_t , the hypermultiplet moduli form a smooth quaternionic Kähler manifold $\mathcal{M}_H^{\text{reg}}$ of quaternionic dimension $h^{2,1}(X_t) + 1$, and the scalar manifold is locally the product of the two. At a root where no Lagrangian description of the light states is available, the Higgs branch is defined through $\mathcal{H}^{\text{reg}}$ alone: it is the metric completion of $\mathcal{H}^{\text{reg}}$ near the limit point, for the metric of $\mathcal{M}_H^{\text{reg}}$ on the hypermultiplet factor and the very special real metric on the vector factor.

3.4 The regularity hypothesis

Hypothesis 3.2 (regularity). Near every point of F and every smooth point of N , the dynamics below the lightest massive state is five-dimensional $N = 1$ gauged supergravity with: the vector multiplets of $\widehat{X}$ with the Wilsonian prepotential (3.1); a smooth quaternionic Kähler manifold M of quaternionic dimension $h^{2,1}(\widehat{X}) + 1 + k$, containing N and the k charged hypermultiplets; and the gauging of the torus T^{k-1} of isometries of M with charge lattice Λ , acting on a neighbourhood of N . No neutral hypermultiplet is charged. Moreover:

1. the Coulomb branch through each root point, M-theory on $\widehat{X}$ with Kähler class near F and vanishing charged hypermultiplets, consists of supersymmetric Minkowski vacua of this theory, and higher-derivative terms do not change the set of such vacua;
2. the Kähler cone of X_t is the same for all smoothings X_t parametrised by a neighbourhood W of each root point x in the hypermultiplet directions, and at every point of $W \cap \mathcal{M}_H^{\text{reg}}$ the vector moduli of the regular Higgs branch form the whole normalised Kähler cone $\mathcal{K}(X_t)$;
3. near each root point, the Higgs vacua of this theory off the root are M-theory on the smoothings X_t , and $\mathcal{M}_H^{\text{reg}}$ is bi-Lipschitz equivalent, near the root, to $(\mu^{-1}(0) \setminus N)/T^{k-1}$ with the quotient metric of [1].

For X_9 the hypothesis concerns M of quaternionic dimension 127, the four charged hypermultiplets of Proposition 2.1 and the torus T^3 with the charges (A.5).

The hypothesis has the form of the standard treatment of conifold singularities. In four dimensions the singularity of the vector-multiplet moduli space at a conifold point is an effect of integrating out the light states: the Wilsonian action that keeps them is smooth [31, 32]. On the hypermultiplet side, for type IIA string theory near k homologous vanishing three-cycles, the metric on the neutral fields alone, corrected by Euclidean D2-branes on the vanishing cycles, has a $\mathbb{C}^2/\mathbb{Z}_k$ singularity [6, 7], the same as the hyperkähler quotient of the Wilsonian description with k light hypermultiplets. The hypothesis asserts the existence of the corresponding smooth Wilsonian hypermultiplet manifold in five dimensions at finite Planck mass, as in the supergravity models of [2]; its metric is not known [2], and nothing below depends on it. The geometric input is proved for X_9 in Proposition 2.1: exactly four hypermultiplets are massless, no divisor collapses, and all gauge couplings are finite. The torus acts on a neighbourhood of N because its weights are integral.

The existence of M and clause (1) are the standard assumptions of the conifold analyses cited above, carried over to finite Planck mass; they are all that Theorem 5.2 uses. Clauses (2) and (3) are technical: they are used only in Lemmas 3.3 and 5.4 and Theorem 5.6, to pass from the four-node roots of X_9 to the other roots through metric completions, which bi-Lipschitz equivalences preserve.

In detail, clause (1) says that the Wilsonian theory reproduces the supersymmetric compactifications of M-theory on $\widehat{X}$, including those with the Kähler class near F , and that the

four-derivative terms do not alter this; it is what forces the moment map to vanish at the root. Clause (2) excludes a dependence of the chamber structure on the complex structure of X_t near the root. Such a dependence does occur in general: the Kähler cone of a Calabi–Yau threefold can jump under deformation of its complex structure, and by Wilson’s theorem this happens only on members containing elliptic quasi-ruled surfaces [33, 34]. So the first half of clause (2) holds if no X_t parametrised by W contains such a surface; the second half, that the vector moduli of the regular Higgs branch fill the whole normalised Kähler cone over W , is an independent assumption. Clause (3) identifies distances measured on the smooth side with those of the Wilsonian quotient up to bounded distortion.

Lemma 3.3. *Over the neighbourhood W of clause (2) of Hypothesis 3.2, the metric of $\mathcal{M}_H^{\text{reg}}$ on the regular Higgs branch does not depend on the vector-multiplet moduli.*

Proof. At every point of $\mathcal{H}^{\text{reg}}$ the compactification is M-theory on a smooth threefold X_t with Kähler class in the interior of its Kähler cone, described by two-derivative supergravity with neutral hypermultiplets. There supersymmetry forbids neutral couplings between the two kinds of multiplets at the two-derivative level, and the scalar manifold is locally the product of the very special real and the quaternionic Kähler factors [35, Sect. 2.4], the form assumed in [29] and the same factorisation that identifies the hypermultiplet moduli spaces of M-theory and of type IIA string theory on one threefold [36, Sect. 2]. So the hypermultiplet metric is locally constant in the vector moduli. By clause (2) of Hypothesis 3.2, over W the regular Higgs branch is, as a set, the product of $W \cap \mathcal{M}_H^{\text{reg}}$ with the whole normalised Kähler cone of X_t ; the latter is identified by radial projection with a convex set, hence connected, and the metric is therefore independent of the vector moduli. The moduli space of vacua does not factorise at transition points themselves [37, Sect. 2.1], but $\mathcal{H}^{\text{reg}}$ contains none. $\square$

4 Quaternionic Kähler quotients near a fixed point

Throughout this section (M, g) is a connected quaternionic Kähler manifold of dimension $4n$, $n \geq 2$, with nonzero reduced scalar curvature $\nu = \text{scal}/(16n(n+2))$, so that $\mathbb{H}\mathbb{P}^n$ with sectional curvature in $[1, 4]$ has $\nu = 1$; in supergravity $\nu < 0$, and M is not assumed compact or complete. Write $\mathcal{Q} \subset \Lambda^2 T^*M$ for the parallel rank-three subbundle spanned locally by $\omega_a = g(I_a \cdot, \cdot)$, $a = 1, 2, 3$, and $\omega = \sum_a \omega_a \otimes \omega_a \in \Gamma(\Lambda^2 T^*M \otimes \mathcal{Q})$. Let G be a connected compact Lie group acting by isometries, with fundamental fields X_ξ , $\xi \in \mathfrak{g}$. Every such isometry preserves $\mathcal{Q}$: for a Killing field X the form $\mathcal{L}_X \Omega$, with $\Omega = \sum_a \omega_a \wedge \omega_a$, is parallel, hence a constant multiple of Ω when the holonomy is $Sp(n)Sp(1)$, and the constant vanishes because $|\Omega|$ is constant; when the holonomy is smaller, M is locally a symmetric quaternionic Kähler space, whose Killing fields preserve $\mathcal{Q}$.

For $\nu \neq 0$ there is a unique section μ_ξ of $\mathcal{Q}$ with $\nabla \mu_\xi = \iota_{X_\xi} \omega$, namely $\mu_\xi = c_0^{-1} (\nabla X_\xi^\flat)^{\text{sp}(1)}$ with c_0 a nonzero multiple of ν fixed by the normalisation of μ , whose value plays no role below [38], [1, Thm. 2.4], [11, p. 434]. Uniqueness holds on every connected open set, since $\mathcal{Q}$ has no nonzero parallel section when $\nu \neq 0$; in particular there are no constants of integration,

and the zero level is canonical [1, pp. 7–8]. The quaternionic Kähler quotient is $\mu^{-1}(0)/G$ with the submersed metric of [1, Thm. 3.1].

4.1 The moment map near a fixed point

Lemma 4.1. *Let p be fixed by G and $A_\xi := \nabla X_\xi(p)$. Then $A_\xi \in \mathfrak{sp}(n) \oplus \mathfrak{sp}(1)_p$ and $\mu_\xi(p)$ is c_0^{-1} times the $\mathfrak{sp}(1)$ -component of A_ξ . The following are equivalent: $\mu(p) = 0$; G acts on $T_p M$ through $Sp(n)$; G acts trivially on $\mathcal{Q}_p$.*

Proof. Since G preserves $\mathcal{Q}$, A_ξ normalises $\mathfrak{sp}(1)_p$. The normaliser of $\mathfrak{sp}(1)$ in $\mathfrak{so}(4n)$ is $\mathfrak{sp}(n) \oplus \mathfrak{sp}(1)$: every derivation of $\mathfrak{sp}(1)$ is inner, and the centraliser of $\mathfrak{sp}(1)$ is $\mathfrak{sp}(n)$. On this subalgebra the projection to $\mathcal{Q}$ is the projection to $\mathfrak{sp}(1)$, because the two ideals are orthogonal for the trace form: on $\mathbb{C}^{2n} \otimes \mathbb{C}^2$ one has $\text{tr}(B \otimes 1 \cdot 1 \otimes C) = \text{tr} B \text{tr} C = 0$. The last equivalence holds because A_ξ acts on $\mathcal{Q}_p \cong \mathfrak{sp}(1)$ by the adjoint action of its $\mathfrak{sp}(1)$ -part and G is connected. $\square$

Lemma 4.1 is essentially in [12, Sect. 4]: the inclusion, the formula for $\mu_\xi(p)$ and the implication from $\mu(p) = 0$ to a trivial action on $\mathcal{Q}_p$ are in the proofs of [12, Prop. 4.2 and Thm. 4.6]; see also [11]. The proof above is included for completeness.

Remark 4.2 (no Fayet–Iliopoulos parameter). On a hyperkähler manifold the moment map of a torus is defined only up to an additive constant $\zeta \in \mathfrak{z}(\mathfrak{g})^* \otimes \mathbb{R}^3$, and the quotients $\mu^{-1}(\zeta)/G$, $\zeta \neq 0$, resolve or deform the singular quotient at $\zeta = 0$, completely or partially; for generic ζ and the flat models of Lemma 5.1 with all $|a_i| = 1$, such as that of X_9 , they are Kronheimer’s asymptotically locally Euclidean spaces [39], and in general they are orbifolds with points of type $\mathbb{Z}_{|a_i|}$. In supersymmetric gauge theory ζ is the Fayet–Iliopoulos parameter. For $\nu \neq 0$ no such constant exists, and Lemma 4.1 identifies the value of the moment map at a fixed point: $\mu(p) \neq 0$ exactly when the isotropy rotates the three complex structures at p . In supergravity language, a nonzero moment map at a fixed point is a gauging of a subgroup of the $SU(2)$ R-symmetry there; for a torus, of a $U(1)$ subgroup. On the Swann bundle, a hyperkähler cone over M for $\nu > 0$ and a pseudo-hyperkähler cone for $\nu < 0$, the quaternionic Kähler quotient corresponds to the hyperkähler quotient at level zero [11]. No other level is possible: the level must be fixed by the $Sp(1)$ acting on the cone, which rotates the three complex structures. So when $\mu(p) = 0$ the singular quotient at $[p]$ cannot be resolved or deformed by a choice of level: its germ is determined by the isotropy representation, as the rest of this section shows.

Lemma 4.3. *Let N be a connected component of M^G . Then $\mu|_N$ is parallel, so $|\mu|$ is constant on N and either $N \subset \mu^{-1}(0)$ or $N \cap \mu^{-1}(0) = \emptyset$. If $\mu|_N = 0$, then N is a totally geodesic quaternionic submanifold with $T_{p'} N = (T_{p'} M)^G$, and over a geodesic ball $N_1 \subset N$ with a frame of $\mathcal{Q}$ parallel along radial geodesics the normal bundle is a $G \times Sp(1)$ -vector bundle whose fibres are mutually isomorphic representations.*

Proof. Along N the fields X_ξ vanish, so $\nabla_Y \mu_\xi = \omega(X_\xi, Y) = 0$ for Y tangent to N . For a fixed point p' the exponential map at p' is G -equivariant, so near p' the fixed set is $\exp_{p'}((T_{p'} M)^G)$,

a totally geodesic submanifold with tangent space the fixed vectors (for the zero set of a single Killing field this is [40]); by Lemma 4.1 the isotropy commutes with the I_a along N , so that space is quaternionic. For the last statement, the multiplicities of the irreducible representations of $G \times Sp(1)$ in the fibres are continuous and integer-valued along the connected base, hence constant. $\square$

Lemma 4.3 is likewise essentially in [12]: for a stabiliser equal to all of G , their Proposition 4.2, with the connectedness step in the proof of their Theorem 4.6, and their Proposition 4.4 give that N is a totally geodesic quaternionic Kähler submanifold, by Gray's theorem on almost quaternionic submanifolds [41], on which μ is parallel; the description of $T_{p'}N$ is in [12, Sect. 2] and the constancy of the normal representation in [12, Sect. 4]. The argument above gives the statement in the form used below.

Let $p \in N$ with $\mu|_N = 0$, let $V = T_p M / T_p N$ be the normal isotropy representation, so that $V^G = 0$, and let

$$\mu_0 : V \rightarrow \mathfrak{g}^* \otimes \mathcal{Q}_p, \quad \mu_0(w)(\xi) = \frac{1}{2} \sum_a g(I_a A_\xi w, w) \omega_a, \quad (4.1)$$

be its flat hyperkähler moment map (with $\mathcal{Q}_p$ identified with $\mathbb{R}^3$ by the frame). By Lemma 4.1 the representation V is quaternionic, and μ_0 is the moment map of the rigid model. We write $S(V)$ for the unit sphere and $L_0 := \mu_0^{-1}(0) \cap S(V)$ for the *link*; since μ_0 is homogeneous of degree two, $\mu_0^{-1}(0)$ is the cone over L_0 .

Lemma 4.4. *For $p' \in N$ and $w \in V_{p'}$, let $\tilde{\mu}(p', w) \in \mathfrak{g}^* \otimes \mathcal{Q}_{p'}$ denote $\mu(\exp_{p'} w)$ in the frame of $\mathcal{Q}$ parallel along $t \mapsto \exp_{p'}(tw)$. Then, uniformly on compact subsets of N ,*

$$\tilde{\mu}(p', w) = \mu_0(w) - \frac{1}{24} \omega(R(A_\xi w, w)w, w) + O(|w|^5). \quad (4.2)$$

In particular there is no cubic term, for every G .

Proof. Let $\gamma(t) = \exp_{p'}(tw)$, $f(t) = \tilde{\mu}(p', tw)$ and $j(t) = X_\xi(\gamma(t))$, a Jacobi field with $j(0) = 0$ and $j'(0) = A_\xi w$. Since ω is parallel and γ is a geodesic,

$$\begin{aligned} f' &= \omega(j, \gamma'), & f'' &= \omega(j', \gamma'), \\ f''' &= -\omega(R(j, \gamma')\gamma', \gamma'), & f'''' &= -\omega((\nabla_{\gamma'} R)(j, \gamma')\gamma' + R(j', \gamma')\gamma', \gamma'). \end{aligned}$$

At $t = 0$: $f(0) = 0$ because $\mu|_N = 0$; $f'(0) = f'''(0) = 0$ because $j(0) = 0$; and $f''(0) = \omega(A_\xi w, w)$, which is symmetric in w because A_ξ commutes with the I_a (Lemma 4.1). Taylor's theorem with smooth dependence on (p', w) gives the expansion; polarisation shows that the whole cubic form vanishes, not only its diagonal. $\square$

4.2 The normal form

Theorem 4.5. *Let (M, g) be a quaternionic Kähler manifold of dimension $4n \geq 8$ with $\nu \neq 0$, not necessarily complete, and G a compact connected group acting isometrically with moment map μ . Let p be a fixed point with $\mu(p) = 0$, N the component of M^G through p , V the normal isotropy representation at p with flat hyperkähler moment map μ_0 , and L_0 its link. Assume that G acts locally freely on $\mu_0^{-1}(0) \setminus \{0\}$. Then:*

1. there exist a neighbourhood N_0 of p in N , $\delta > 0$, a G -invariant neighbourhood U of p with $U \cap M^G = N_0$, and a G -equivariant homeomorphism

$$\Psi : N_0 \times (\mu_0^{-1}(0) \cap B_\delta(V)) \longrightarrow \mu^{-1}(0) \cap U, \quad \Psi(p', 0) = p', \quad (4.3)$$

which is a diffeomorphism away from $N_0 \times \{0\}$; hence the germ of the quaternionic Kähler quotient at $[p]$ is that of $N \times (V///G)$, compatibly with orbit types;

2. if the action is free off the origin, $(\mu^{-1}(0) \cap U \setminus N_0)/G$ is a smooth quaternionic Kähler manifold of dimension $\dim M - 4 \dim G$;
3. Ψ can be chosen of the form $\Psi(p', r\hat{w}) = \exp_{p'}(\sigma(p') r \tau(p', r, \hat{w}))$, where σ is a smooth equivariant isometry from the model V onto the normal spaces, parallel along radial geodesics of N , and $\tau(p', r, \cdot) : L_0 \rightarrow S(V)$ is smooth up to $r = 0$ with $\tau(p', 0, \cdot) = \text{id}$, $\tau - \text{id} = O(r^2)$, $D_{(p', \hat{w})}(\tau - \text{id}) = O(r^2)$ and $\partial_r \tau = O(r)$, uniformly on N_0 .

In words: near a fixed component with vanishing moment map, the zero set of μ is a bundle over N whose fibre is the zero set of the rigid model, bent by an amount of order r^2 at distance r from N . The proof has three steps. The first sets up coordinates normal to N . The second rescales the moment map by r^{-2} and shows that the rescaled zero set is a smooth manifold transverse to the link at $r = 0$. The third moves the link along r by the flow of a connection; the vanishing of the cubic term in (4.2) makes the flow stationary to first order at $r = 0$, which gives the rate r^2 .

Proof. Step 1: coordinates. Over a geodesic ball $N_1 \ni p$ with a radially parallel frame of $\mathcal{Q}$, Lemma 4.3 provides σ ; in these coordinates μ_0 does not depend on the base point. Choose δ below the normal injectivity radius over N_1 ; then normal coordinates on $U = \exp(T^\perp N|_{N_0} \cap B_\delta)$ are unique, and $U \cap M^G = N_0$ because a fixed point $\exp_{p'}(w)$ forces $gw = w$ for all g , hence $w = 0$ as $V^G = 0$.

Step 2: the rescaled zero set. Put $\Phi(p', r, \hat{w}) := r^{-2} \tilde{\mu}(p', r\hat{w})$ for $\hat{w} \in S(V)$ and $0 < r < \delta$. By Lemma 4.4, $\Phi = \mu_0(\hat{w}) + O(r^2)$, so Φ extends smoothly to $r = 0$ with $\Phi(p', 0, \cdot) = \mu_0$, $\partial_{p'} \Phi = 0$ and $\partial_r \Phi = 0$ there. We show that the differential of μ_0 in the $\hat{w}$ -direction is onto at every point of L_0 . Its adjoint is $(\xi_a)_a \mapsto \sum_a I_a X_{\xi_a}(\hat{w})$. On the zero level the three subspaces $I_a \mathfrak{g} \cdot \hat{w}$ are mutually orthogonal, since $g(I_a X_\xi, I_b X_\eta) = \pm \omega_c(X_\xi, X_\eta) = \pm \mu_0^c(\hat{w})([\xi, \eta]) = 0$, and each is the image of $\mathfrak{g} \cdot \hat{w}$ under an isometry; local freeness makes $\xi \mapsto X_\xi(\hat{w})$ injective, so the adjoint is injective and $d\mu_0$ is onto. By Euler's identity $d\mu_0(\hat{w})\hat{w} = 2\mu_0(\hat{w}) = 0$, so the restriction to $T_{\hat{w}}S(V)$ is still onto. Choose a neighbourhood O of L_0 in $S(V)$ on which $\partial_{\hat{w}} \Phi$ stays onto for small r , uniformly on N_0 by compactness. Outside O , $|\mu_0| \geq c > 0$, while $|\Phi - \mu_0| \leq Cr^2$; so for $C\delta^2 < c$ the function Φ has no zeros outside O . Hence its zero set

$$Z := \{(p', r, \hat{w}) \in N_0 \times [0, \delta) \times S(V) : \Phi(p', r, \hat{w}) = 0\}$$

is a closed submanifold with boundary $Z \cap \{r = 0\} = N_0 \times \{0\} \times L_0$, and the projection $\pi : Z \rightarrow N_0 \times [0, \delta)$ is a proper G -equivariant submersion.

Step 3: trivialisation. Give $N_0 \times [0, \delta) \times S(V)$ the product metric $g_N \oplus dr^2 \oplus g_{S(V)}$, and on Z take the connection whose horizontal spaces are the orthogonal complements of the fibres of π ; it is G -invariant, since G acts isometrically on $S(V)$, trivially on the other factors, and preserves Z . Let $\tilde{\partial}_r$ be the horizontal lift of ∂_r ; it has no N_0 -component and r -component 1. At $r = 0$ the vector $(0, 1, 0)$ is tangent to Z , because $\partial_r \Phi = 0$ there, and it is orthogonal to the fibre $N_0 \times \{0\} \times L_0$; so $\tilde{\partial}_r = (0, 1, 0)$ at $r = 0$. Its $S(V)$ -component Y_S is therefore a smooth function vanishing at $r = 0$, and Y_S and its derivatives in p' and $\hat{w}$ are $O(r)$. Let $\tau(p', r, \hat{w})$ be the $S(V)$ -component of the flow of $\tilde{\partial}_r$ at time r from $(p', 0, \hat{w})$, $\hat{w} \in L_0$. Then $\partial_r \tau = Y_S(p', r, \tau) = O(r)$ gives $\tau - \text{id} = O(r^2)$, and differentiating the flow equation in $(p', \hat{w})$ and applying Gronwall's inequality gives the same bound for the first derivatives; this is (3). If the expansion (4.2) had a cubic term, $\partial_r \Phi$ would not vanish at $r = 0$, Y_S would be of order one there, and the flow would only give $\tau - \text{id} = O(r)$.

The map Ψ of (3) is a G -equivariant bijection onto $\mu^{-1}(0) \cap U$, since $N_0 \subset \mu^{-1}(0)$ by Lemma 4.3; it is a diffeomorphism for $r > 0$ and continuous with continuous inverse along N_0 . Being G -equivariant, it preserves stabilisers and descends to a homeomorphism of orbit spaces that preserves orbit types. This proves (1). For (2), $\partial_{\hat{w}} \Phi$ is onto on Z and $d\tilde{\mu} = r^2 d\Phi$ for $r > 0$, so μ is a submersion on $\mu^{-1}(0) \setminus N$, and [1, Thm. 3.1] applies; (2) is also the case of trivial stabiliser of [12, Prop. 4.4]. $\square$

The topological type of the germ therefore depends only on the isotropy representation: neither ν nor the hyperkähler part of the curvature nor any higher jet of the metric enters.

4.3 The quotient metric and its tangent cone

Let $Z_M := \mu^{-1}(0)$ carry the intrinsic length metric d_{Z_M} induced by g , and let $M_{\text{red}} := Z_M/G$ carry the quotient metric $d_{\text{red}}([x], [y]) = \inf_{h \in G} d_{Z_M}(hx, y)$. Where G acts freely, horizontal lifts preserve length, so off the image of N the metric d_{red} is locally the Riemannian distance of the Galicki–Lawson metric [1]. For compact connected M and G this is the length structure of Dancer and Swann, who show that each component of M_{red} is then a complete length space whose topology is the quotient topology [12, Sect. 7, Thm. 7.1]. The statements below are local and assume neither compactness nor completeness.

Lemma 4.6. *In the situation of Theorem 4.5, assume that the link L_0 is nonempty and connected. Then near $[p]$ the intrinsic metrics of Z_M and of $Z_M \setminus N$ agree on $Z_M \setminus N$, and $(M_{\text{red}}, d_{\text{red}})$ is near $[p]$ the metric completion of its part off the image of N . The assumption holds for a torus with $\mu_0^{-1}(0) \neq \{0\}$.*

Proof. For a torus acting on $V = \mathbb{H}^k$ with weights q , write μ_i for the moment map of the i -th summand. Then L_0 is the preimage under the map $w \mapsto (\mu_i(w))_i$ of the set

$$\left\{ (\mu_i)_i \in (\mathbb{R}^3)^k : \sum_i q_i \otimes \mu_i = 0 \text{ for all } q, \sum_i |\mu_i| = \frac{1}{2} \right\},$$

the unit sphere of a norm on the subspace $K \otimes \mathbb{R}^3$, where $K \neq 0$ is the annihilator of the weights in $\mathbb{R}^k$; this subspace has dimension at least three, so the sphere is connected. The map is proper and onto, with connected fibres (products of circles and points), so L_0 is connected.

Work in the chart Ψ of Theorem 4.5(3), with coordinates $(v, r, \hat{w})$ on $Z_M \cap U$, and compare the induced metric with $g_0 = g_N \oplus dr^2 \oplus r^2 g_{L_0}$. At $r = 0$ the differential of $\exp$ is an isometry along the normal directions, and the variation along N is a Jacobi field with initial value the variation of v and initial derivative given by the normal connection applied to σw , which is of order r ; since τ is smooth up to $r = 0$, the pullback of g is $(1 + O(r))g_0$ uniformly on N_0 . Let γ be a curve in $Z_M \cap U$ of length ℓ joining two points off N , and let $0 < \eta' < \eta$ be smaller than the values of r at its endpoints. Consider the components I of $\{t : r(\gamma(t)) < \eta\}$ that contain a zero of r . On each, γ descends from $r = \eta$ to $r = 0$ and returns, so its length there is at least η , and there are at most $\ell/\eta + 1$ such components. In such a component let a and b be the first and last zeros of r , let $a' < a$ be the last time before a with $r = \eta'$ and $b' > b$ the first time after b with $r = \eta'$. Replace $\gamma|_{[a', b']}$ by the curve that moves along L_0 at radius η' from $\gamma(a')$ to a fixed point $(v(a'), \eta', \hat{w}_*)$, then follows $t \mapsto (v(t), \eta', \hat{w}_*)$, and moves along L_0 at radius η' to $\gamma(b')$; this uses that L_0 is connected. The new curve avoids N . Its middle piece is at most $(1 + O(\eta))$ times as long as $\gamma|_{[a', b']}$, since the g_0 -length of a curve bounds the g_N -length of its projection, and the two arcs in L_0 have length $O(\eta')$. The total added length is therefore at most $(\ell/\eta + 1)O(\eta') + O(\eta)\ell$, which tends to zero when first $\eta' \rightarrow 0$ and then $\eta \rightarrow 0$. So the two intrinsic metrics agree on $Z_M \setminus N$.

The same comparison with g_0 , together with the connectedness of L_0 , shows that near N the intrinsic metric of Z_M is bounded by a constant times the ambient distance, so $Z_M \cap \bar{U}$ is complete for the intrinsic metric. Since $Z_M \setminus N$ is dense in Z_M , Z_M is near p the completion of $Z_M \setminus N$. Passing to orbit metrics commutes with completion for a compact group acting isometrically, which gives the statement for M_{red} . $\square$

Proposition 4.7. *In the situation of Theorem 4.5, with the action free off the origin, the pointed rescalings $(M_{\text{red}}, \varepsilon^{-1}d_{\text{red}}, [p])$ converge in the pointed Gromov–Hausdorff sense to $T_p N \times V///G$ with its flat product metric. More precisely, for every $R > 0$ and small $\varepsilon > 0$ there is a homeomorphism from the ball of radius R about the vertex of $T_p N \times V///G$ onto a neighbourhood of $[p]$ in $(M_{\text{red}}, \varepsilon^{-1}d_{\text{red}})$, sending the vertex to $[p]$, that is bi-Lipschitz onto its image with constant $1 + O(\varepsilon^2 R^2)$; its image contains the ball of radius $R(1 - C\varepsilon^2 R^2)$ about $[p]$. In particular, at distance r from $[p]$ the metric differs from the flat model by relative corrections of order r^2 .*

Proof. Use the coordinates $x = (v, w) \mapsto \exp_{\exp_p^N(v)}(\sigma(\exp_p^N v)w)$ of Theorem 4.5(3); since σ and the frame of $\mathcal{Q}$ are radially parallel, these are Fermi coordinates along the totally geodesic N , and $g = g_p + O(|x|^2)$. Rescale: $g_\varepsilon = \varepsilon^{-2}(\text{pullback of } g \text{ under } x \mapsto \varepsilon x)$ satisfies $g_\varepsilon = g_p + O(\varepsilon^2 R^2)$ on the ball of radius $3R$. The rescaled chart $\Psi_\varepsilon(v, r\hat{w}) = (v, r\tau(\exp^N(\varepsilon v), \varepsilon r, \hat{w}))$ maps $T_p N \times \text{cone}(L_0)$ onto the rescaled zero set, and by Theorem 4.5(3) it differs from the inclusion by $O(\varepsilon^2 R^2)$ in C^1 there. It preserves v and $|w|$ and is the identity on $\{r = 0\}$. Since $|\Psi_\varepsilon - \text{id}| = r|\tau - \text{id}| = O(\varepsilon^2 R^2)r$ and $d\Psi_\varepsilon - \text{id} = O(\varepsilon^2 R^2)$ for $r > 0$, it is Lipschitz in the chart with constant $1 + O(\varepsilon^2 R^2)$.

Let γ be a rectifiable curve in the model cone. Then $\Psi_\varepsilon \circ \gamma$ is rectifiable and differentiable almost everywhere. At almost every time either $r(\gamma(t)) > 0$, where the C^1 estimate compares

the speeds of γ and $\Psi_\varepsilon \circ \gamma$, or $\gamma(t)$ lies in $\{r = 0\}$ at a density point of the set of such times (the other times form a null set, by the Lebesgue density theorem), where $\gamma'(t)$ is tangent to $\{r = 0\} = T_p N$ and Ψ_ε acts as the identity; there the two metrics differ by $O(\varepsilon^2 R^2)$. Integrating, the g_ε -length of $\Psi_\varepsilon \circ \gamma$ is $1 + O(\varepsilon^2 R^2)$ times the flat length of γ . The same holds for Ψ_ε^{-1} , and near-minimising curves between points of the ball of radius R stay in the ball of radius $3R$. The intrinsic metrics are therefore $(1 + O(\varepsilon^2 R^2))$ -bi-Lipschitz, and since Ψ_ε is equivariant so are the orbit metrics. Since Ψ_ε is defined on the ball of radius $3R$ and small d_{red} -balls about $[p]$ lie in the chart, every point at rescaled distance less than $R(1 - C\varepsilon^2 R^2)$ from $[p]$ is the image of a model point of norm less than R . $\square$

For the transition of Section 2, under Hypothesis 3.2, the model is $V = \mathbb{H}^4$ with the weights (A.5) and $V///T^3 = \mathbb{C}^2/\mathbb{Z}_4$ with its flat orbifold metric: right multiplication by j on a hypermultiplet reverses the sign of its charge and preserves the hyperkähler structure, which carries the effective torus to that of Kronheimer's A_3 quotient at zero level [39]. The leading behaviour at the root is therefore the flat cone over the round $S^3/\mathbb{Z}_4$, and the order of the cyclic group is fixed by the charges.

Remark 4.8. The following examples illustrate Theorem 4.5 and Proposition 4.7.

1. On quaternionic projective space $\mathbb{HP}^d$, normalised to sectional curvature in $[1, 4]$, with a torus acting linearly in homogeneous coordinates, fixing a point p and acting trivially on its homogeneous coordinate (equivalently $\mu(p) = 0$), the moment map in the radially parallel frame is exactly $\mu_0(w) (\sin |w|/|w|)^2$ for $|w| < \pi/2$; on the noncompact dual it is $\mu_0(w) (\sinh |w|/|w|)^2$. The zero set is then exactly the flat cone in normal coordinates, and the expansion (4.2) can be read off.
2. On quaternionic hyperbolic space of quaternionic dimension four, normalised to sectional curvature in $[-4, -1]$, with T^3 acting linearly at the origin with the weights (A.5) (so N is a point), the quotient metric is exactly $dr^2 + \frac{1}{4} \sinh^2(2r) g_{S^3/\mathbb{Z}_4}$, real hyperbolic space of curvature -4 divided by $\mathbb{Z}_4$. Since $\frac{1}{4} \sinh^2 2r = r^2(1 + \frac{4}{3}r^2 + O(r^4))$, the coefficient of the link metric deviates from the flat one at relative order $\frac{4}{3}r^2$, and the bi-Lipschitz constant of Proposition 4.7 is $1 + \frac{2}{3}R^2 + O(R^4)$: the rate is attained. The same torus acting on four homogeneous coordinates of $\mathbb{HP}^d$, $d \geq 5$, gives a quotient whose germ along $N = \mathbb{HP}^{d-4}$ is $N \times \mathbb{C}^2/\mathbb{Z}_4$, with slice metric $d\rho^2 + \frac{1}{4} \sin^2(2\rho) g_{S^3/\mathbb{Z}_4}$.
3. The hypothesis $\mu(p) = 0$ cannot be dropped: the circle acting on $\mathbb{HP}^d$ by left multiplication with $e^{i\theta}$ has fixed set $\mathbb{CP}^d$, on which $|\mu|$ is a nonzero constant, and its quotient $\text{Gr}_2(\mathbb{C}^{d+1})$ does not meet the image of the fixed set.

Remark 4.9 (hyperkähler and Kähler quotients). The proofs of Lemmas 4.3 and 4.4, Theorem 4.5, Lemma 4.6 and Proposition 4.7 use the quaternionic Kähler structure only through three facts: $\nabla \mu_\xi = \iota_{X_\xi} \omega$ with ω parallel, $\mu(p) = 0$, and the action of the isotropy at p through a group preserving the flat structure defined by the ω_a on $T_p M$. They therefore hold verbatim for a Riemannian hyperkähler manifold, not necessarily complete, with a compact

connected group acting tri-Hamiltonianly, at any fixed point p : for an equivariant moment map μ , $\mu' := \mu - \mu(p)$ is again equivariant, since $\mu(p) = \mu(gp) = \text{Ad}_g^* \mu(p)$, so that the statements concern $\mu^{-1}(\mu(p))/G$. Moreover, the isotropy commutes with the complex structures at every fixed point, the frame of $\mathcal{Q}$ is globally parallel, and part (2) of Theorem 4.5 uses the hyperkähler quotient construction [42] on the open set where the action is free. Theorem 4.5, Proposition 4.7 and Lemma 4.6, under its hypothesis that the link is connected, also hold for Kähler quotients at a fixed point, at the level $\mu(p)$ after the same shift, with the flat Kähler quotient of the isotropy representation as the model; there part (1) of Theorem 4.5 is contained in the symplectic local normal form [13, 14] and its consequences for singular reduction [15], and the metric statements are what the proofs add. For positive-definite hyperkähler manifolds on which the action of the complexified group exists, that is, the vector fields $I_1 X_\xi$ are complete, a complex-analytic local model at any point of the quotient is due to Mayrand [17, Thm. 1.4]; its proof uses the global Kähler quotient and does not apply to an invariant neighbourhood of p alone. The metric tangent cone of the quotient metric, with the rate r^2 , we have not found in the literature; tangent cones are not considered in the length structure of [12, Sect. 7] recalled in Section 4.3. The closest statements are the expansion $dr^2 + r^2(g_0 + O(r^2))$ of the L^2 metric near a reducible instanton [43, Thm. I; cf. Thm. 2.8], the product structure, up to a term vanishing with the distance to the stratum, at points of the singular strata of flat hyperkähler cones of quiver data away from the vertex [44], and the description of the tangent cone of a Riemannian orbit space M/G at $[p]$ as the normal space divided by the isotropy [45, Sect. 2], which concerns the extrinsic metric of $\mu^{-1}(0)/G$ rather than its length metric. When M is a hyperkähler or quaternionic Kähler four-manifold the link L_0 is empty and these statements are vacuous. Lemma 4.10 and Proposition 4.12 below also hold for hyperkähler quotients, with $\nu = 0$ and $R = R_{\text{hk}}$.

4.4 The leading correction

Proposition 4.7 says that the quotient metric differs from the flat cone at relative order r^2 . We now determine this correction in the direction transverse to N . By Alekseevsky's theorem [46] the curvature of M at p is

$$R = \nu R_{\text{HIP}} + R_{\text{hk}}, \quad (4.4)$$

where R_{HIP} is the curvature tensor of HIP^n with sectional curvatures in $[1, 4]$ and R_{hk} is an algebraic curvature tensor of hyperkähler type, that is, commuting with the I_a . In supergravity ν is fixed in Planck units, while R_{hk} depends on the compactification.

In the situation of Theorem 4.5, the *transverse slice* at p , which realises the transverse type of Section 1 metrically, is $\mathcal{Y}_p = (\mu^{-1}(0) \cap \exp_p(V \cap B_\delta))/G$ with the length metric induced from M_{red} ; by Theorem 4.5 its germ at $[p]$ is that of $V///G$.

Lemma 4.10. *In normal coordinates at p , the zero set of μ in $\exp_p(V)$ is $\{x + \psi(x)\}$ with x in the zero set $\mu_0^{-1}(0)$ of the flat moment map, where $\psi(x)$ lies in the span of the vectors $I_b A_\eta x$ of the flat model, is unique in a conic neighbourhood $|\psi| \leq \varepsilon|x|$, and satisfies $\psi = \psi_3 + O(|x|^4)$ with ψ_3 homogeneous of degree three.*

Proof. Write $x = r\hat{x}$ and $\psi = r\varphi$. On the link, $d\mu_0(\hat{x})$ restricted to the span of the $I_b X_\eta(\hat{x})$ is an isomorphism (Step 2 of the proof of Theorem 4.5), so the implicit function theorem applied to $r^{-2}\tilde{\mu}(r(\hat{x} + \varphi)) = 0$ at $(r, \varphi) = (0, 0)$, uniformly over the compact link, gives φ ; by (4.2) $\varphi = O(r^2)$, since there is no cubic term, so ψ has a cubic leading part, and the quintic term of the expansion gives a remainder of order four. $\square$

We identify $\mathbb{H} = \mathbb{C} \oplus \mathbb{C}j$ with $\mathbb{C}^2$ by $(\alpha, \beta) \mapsto \alpha + \beta j$, with I_1, I_2, I_3 acting by left multiplication by i, j, k , and orient $\mathbb{C}^2$ so that the Kähler forms of the flat structure are self-dual; tensors of hyperkähler type on $\mathbb{C}^2$ are then the anti-self-dual Weyl tensors, and form a real space of dimension five. (In the opposite orientation, the usual one for four-dimensional quaternionic Kähler geometry, these are self-dual.)

Definition 4.11. Suppose that $V///G$ has real dimension four, that G acts freely on $\mu_0^{-1}(0) \setminus \{0\}$, and that there is a linear map $\ell : \mathbb{H} \rightarrow V$ commuting with the I_a , with $\ell(\mathbb{H}) \subset \mu_0^{-1}(0)$ and horizontal, $\ell(\mathbb{H}) \perp \mathfrak{g} \cdot \ell(y)$ for every y . Then $|\ell y| = c_\ell |y|$ for a constant c_ℓ ; let $\Gamma \subset G$ be the stabiliser of the quaternionic line $\ell(\mathbb{H})$. The *uniformising chart determined by ℓ* is $y \mapsto [\exp_p(\ell y + \psi(\ell y))]$, from a neighbourhood of 0 in $\mathbb{H}$ to $\mathcal{Y}_p$, with ψ as in Lemma 4.10. The pulled-back metric $g^\mathcal{Y}$ is not known to be twice differentiable at 0. If, in the coordinate $z = c_\ell y$, it has an expansion $g_{ij}^\mathcal{Y}(z) = \delta_{ij} + Q_{ij}(z) + O(|z|^3)$ with Q quadratic, the *leading curvature* R_0 of $\mathcal{Y}_p$ is the curvature at 0 of the metric $\delta + Q$; Proposition 4.12 shows that such an expansion exists. Where $g^\mathcal{Y}$ is twice differentiable at 0, R_0 is its curvature there.

For the torus models of Lemma 5.1, $\ell(\alpha, \beta)$ is $\sqrt{|a_i|}(\alpha, \beta)$ on the hypermultiplets with $a_i > 0$ and $\sqrt{|a_i|}(\beta, -\alpha)$ on those with $a_i < 0$, in the coordinates $(z_i, \bar{z}_i)$ of the proof of that lemma; then $c_\ell^2 = m$ and $\Gamma = \mathbb{Z}_m$. For these models any other map satisfying the conditions of Definition 4.11 is $h \circ \ell \circ (s\rho_v)$ with $h \in G$, $s > 0$ and ρ_v right multiplication by a unit quaternion, an isometry of $\mathbb{H}$ commuting with the I_a ; so the line $\ell(\mathbb{H})$ is unique up to G , and R_0 is defined up to the action of the right $Sp(1)$, which acts on the anti-self-dual Weyl tensors and conjugates Γ . By Proposition 4.12, R_0 is a Γ -invariant algebraic curvature tensor, and the pulled-back metric is $|dz|^2 - \frac{1}{3}\langle R_0(\cdot, z)z, \cdot \rangle + O(|z|^3)$.

Proposition 4.12. Assume that $V///G$ has real dimension four, that G acts freely on $\mu_0^{-1}(0) \setminus \{0\}$, that the link L_0 is connected (automatic for a torus, Lemma 4.6), and that a map ℓ as above exists. Then the leading curvature of $\mathcal{Y}_p$ (Definition 4.11) is the curvature of M at p restricted to the quaternionic line $\ell(\mathbb{H})$:

$$R_0 = c_\ell^{-4} \ell^*(R|_{\ell(\mathbb{H})}). \quad (4.5)$$

Consequently $\nu R_{\mathbb{H}\mathbb{P}}$ contributes $4\nu R^{(1)}$, where $R^{(1)}(X, Y)Z = \langle Y, Z \rangle X - \langle X, Z \rangle Y$ is the algebraic curvature tensor of constant sectional curvature 1 on $\mathbb{H}$, and R_{hk} contributes the anti-self-dual Weyl tensor $c_\ell^{-4} \ell^*(R_{\text{hk}}|_{\ell(\mathbb{H})})$. If the latter vanishes, $\mathcal{Y}_p$ agrees to second order at its vertex with the quotient by Γ of the space form of curvature 4ν .

Proof. First, ℓ induces a homothety onto $V///G$: the induced map from the unit sphere $S^3 \subset \mathbb{H}$ to L_0/G is a local homothety between compact connected three-manifolds, hence a covering and onto, and if $h\ell(y) = \ell(y')$, $h \in G$, then $h\ell(\mathbb{H})$ and $\ell(\mathbb{H})$ are both the horizontal space of the zero set at $\ell(y')$, so $h \in \Gamma$. In normal coordinates at p the metric is $g_x(Y, Z) = \langle Y, Z \rangle - \frac{1}{3} \langle R(Y, x)x, Z \rangle + O(|x|^3)$. The tangent vectors of the surface $y \mapsto \ell y + \psi(\ell y)$ of Lemma 4.10 are $\ell v + \partial_v \psi$. Their pairings $\langle \ell v, \partial_w \psi \rangle$ vanish: since ℓ commutes with the I_b and its image is orthogonal to the orbits, $\langle \ell v, I_b X_\eta(\ell y) \rangle = 0$ for all v, y , and differentiating in y gives the claim. The remainder $\psi - \psi_3 = O(|x|^4)$ changes the metric by $O(r^5)$. The components of the tangent vectors along the orbits are $O(|R|r^3)$, against an orbit Gram matrix of size r^2 , so passing to the horizontal part changes the metric by $O(|R|^2 r^4)$. Hence the metric in the uniformising chart is $c_\ell^2 |dy|^2 - \frac{1}{3} \langle R(\ell \cdot, \ell y) \ell y, \ell \cdot \rangle + O(|y|^3)$, which is (4.5) after rescaling. For Y in the line spanned by $x, I_1 x, I_2 x, I_3 x$ one has $\langle R_{\mathbb{H}\mathbb{P}}(Y, x)x, Y \rangle = |x|^2 |Y|^2 - \langle x, Y \rangle^2 + 3 \sum_b \langle I_b x, Y \rangle^2 = 4(|x|^2 |Y|^2 - \langle x, Y \rangle^2)$: quaternionic lines have sectional curvature 4. A tensor of hyperkähler type restricted to a quaternionic line is of hyperkähler type in dimension four, hence an anti-self-dual Weyl tensor. $\square$

When N is a point the quotient is a four-dimensional quaternionic Kähler orbifold with the same ν [1], that is, Einstein with one half of its Weyl tensor vanishing (with the orientation above, the self-dual half), so the form of its curvature at smooth points is forced. The content of Proposition 4.12 is the formula (4.5) for the leading curvature at the vertex. For $\nu = -1$ and $R_{\text{hk}} = 0$ it is consistent with the metric $dr^2 + \frac{1}{4} \sinh^2(2r) g_{S^3/\mathbb{Z}_4}$ of Remark 4.8(2), whose relative correction $\frac{4}{3}r^2$ is that of curvature -4 . In four-dimensional $N = 2$ supergravity, in units $\kappa^2 = 1$ and for the metric of the hypermultiplet manifold normalised by the Lagrangian of [5], its scalar curvature is $-8n_H(n_H + 2)$ [5, eq. (2.3)], so that $\nu = -\frac{1}{2}$ and the universal part of the leading curvature is $-2R^{(1)}$; the five-dimensional value, in units of the five-dimensional Planck mass, is not needed here.

5 The Higgs branch at finite Planck mass

5.1 Faces with one relation

In this subsection $\widehat{X}$ and F satisfy (S1)–(S3) of Section 3.1. The argument uses only the charges of the light states. Its flat model is the rigid model of the root.

Lemma 5.1. *Let T^{k-1} act on $V = \mathbb{H}^k$ with charge lattice Λ as in (S2), and put $m = \sum_i |a_i|$. Then T^{k-1} acts freely on $\mu_0^{-1}(0) \setminus \{0\}$, and $V///T^{k-1}$ is $\mathbb{C}^2/\mathbb{Z}_m$ with its flat orbifold metric; its coordinate ring is $\mathbb{C}[X, Y, s]/(XY - c s^m)$ for a nonzero constant c .*

Proof. Write $\mu_i \in \mathbb{R}^3$ for the hyperkähler moment map of the i -th summand $\mathbb{H}$, so that $|\mu_i| = \frac{1}{2}|w_i|^2$ and $\mu_0 = \sum_i q_i \otimes \mu_i$ on the charge vectors $q \in \Lambda$. If $\mu_0(w) = 0$, each of the three components of $(\mu_1, \dots, \mu_k)$ annihilates Λ , hence is a multiple of a . If some $w_i = 0$ then $\mu_i = 0$, and since $a_i \neq 0$ the multiples vanish, so $w = 0$. Every nonzero point of $\mu_0^{-1}(0)$ therefore has all coordinates nonzero. Write Q for a $(k-1) \times k$ matrix whose rows are a basis

of Λ , and let $\theta \in \mathbb{R}^{k-1}/\mathbb{Z}^{k-1}$ act on the i -th summand with weight $Q^\top \theta$. The stabiliser of a point with all coordinates nonzero is $\{\theta : Q^\top \theta \in \mathbb{Z}^k\}$, which is trivial because Λ is saturated, that is because the Smith invariants of Q are all 1.

In one complex structure write $w_i = (z_i, \tilde{z}_i)$, with z_i of charge q_i and $\tilde{z}_i$ of charge $-q_i$, and put $u_i = z_i \tilde{z}_i$. A monomial in the $z_i, \tilde{z}_i$ is invariant if and only if the difference of its exponent vectors lies in $\mathbb{Z}a$, so the invariant ring of $\mathbb{C}[z, \tilde{z}]$ is generated by $u_1, \dots, u_k$ and

$$X = \prod_{a_i > 0} z_i^{a_i} \prod_{a_i < 0} \tilde{z}_i^{-a_i}, \quad Y = \prod_{a_i > 0} \tilde{z}_i^{a_i} \prod_{a_i < 0} z_i^{-a_i},$$

and every invariant monomial has a unique normal form $u^\gamma X^n$ or $u^\gamma Y^n$, so the single relation $XY = \prod_i u_i^{|a_i|}$ generates all relations. The components of the complex moment map are the invariant linear forms $\sum_i q_i u_i$, $q \in \Lambda$, so by the Reynolds operator the ideal they generate meets the invariant ring in the ideal they generate there; on their zero set $u = s a$ for a scalar s . The invariant ring of the complex level set is therefore $\mathbb{C}[X, Y, s]/(XY - c s^m)$, where $c = \prod_i \text{sign}(a_i)^{|a_i|} |a_i|^{|a_i|}$, a normal domain: the A_{m-1} surface singularity. By the Kempf–Ness theorem [47], $V///T^{k-1}$ is homeomorphic, and biholomorphic in the chosen complex structure, to the spectrum of this ring.

Finally, $V///T^{k-1}$ is a hyperkähler cone of real dimension four, smooth off its vertex. A Ricci-flat cone over a three-manifold has an Einstein link, which in dimension three has constant curvature, so the cone is flat, $\mathbb{C}^2/\Gamma$ with $\Gamma \subset SU(2)$, and the coordinate ring identifies $\Gamma = \mathbb{Z}_m$. $\square$

Theorem 5.2. *Let $\hat{X}$ and F satisfy (S1)–(S3), and assume Hypothesis 3.2. Let $x = (\phi, b, 0)$ be a root point with $\phi \in F$ and b a smooth point of N . Then, as germs of stratified spaces,*

$$\mathcal{V}_x \cong (\mathcal{K}_\phi \times N) \cup (F_\phi \times N \times \mathbb{C}^2/\mathbb{Z}_m), \quad m = \sum_i |a_i|, \quad (5.1)$$

the Coulomb branch and the Higgs branch, glued along $F_\phi \times N \times \{0\}$, where N stands for its germ at b . The hypermultiplet factor of the Higgs branch off its root is a smooth quaternionic Kähler manifold of quaternionic dimension $h^{2,1}(\hat{X}) + 2$, there is no mixed branch, and the metric of the hypermultiplet factor is the flat orbifold metric of $\mathbb{C}^2/\mathbb{Z}_m$ times the flat metric on $T_b N$ up to relative corrections of order r^2 , where r is the distance from the root.

Proof. By clause (1) of Hypothesis 3.2 the Coulomb branch through x , on which the charged hypermultiplets vanish, consists of supersymmetric Minkowski vacua. By (S3) the vector kinetic matrix is nondegenerate at ϕ , so by (3.3) the moment maps vanish there; moreover, $\mu|_N$ is parallel (Lemma 4.3) and vanishes at the nearby Coulomb points, hence on N . So $\mu(p) = 0$ at the point $p \in M$ over b with vanishing charged hypermultiplets. Since no neutral hypermultiplet is charged, N is the component through p of the fixed set of T^{k-1} in M , which is the N of Section 4. By Lemma 4.1, T^{k-1} acts on $T_p M$ through $Sp(n)$, trivially on $T_p N$ and with charge lattice Λ on $V = \mathbb{H}^k$. By Lemma 5.1 the action is free on $\mu_0^{-1}(0) \setminus \{0\}$, so Theorem 4.5 gives the hypermultiplet germ $N \times (V///T^{k-1}) = N \times \mathbb{C}^2/\mathbb{Z}_m$, the part off N being

a smooth quaternionic Kähler manifold of quaternionic dimension $(h^{2,1}(\widehat{X}) + 1) + k - (k - 1)$. The metric statement is Proposition 4.7 together with the flatness in Lemma 5.1.

The vector conditions are the last equation of (3.3). Off the root of the Higgs branch T^{k-1} acts freely, so $h^I k_I = 0$ forces the image of h in the Lie algebra of T^{k-1} to vanish, that is $h \cdot C_i = 0$ for all i ; since F is open in the annihilator of the C_i by (S1), the vector moduli lie in F_ϕ . On the Coulomb branch every $h \in \mathcal{K}_\phi$ is allowed. By Theorem 4.5 the zero set near p is modelled on $N \times \mu_0^{-1}(0)$, whose points off N have all charged coordinates nonzero, so there is no mixed branch. $\square$

When all $a_i = \pm 1$, the case of k nodes whose vanishing three-spheres are homologous up to sign, $m = k$ and the transverse type is $\mathbb{C}^2/\mathbb{Z}_k$. This is the configuration for which Greene, Morrison and Vafa found $\mathbb{C}^2/\mathbb{Z}_k$ with gravity decoupled [10, Sect. 3, eqs. (3.13)–(3.18)] and expected its qualitative features to persist in the four-dimensional hypermultiplet space [10, Sect. 6]. The germ in Theorem 5.2 is the rigid model of the k light hypermultiplets: gravity does not remove the singular point or change its type.

5.2 The four-node roots of X_9

We call the roots over the face F of (2.3), where $\widehat{X}_9$ contracts to the threefold X_F with four nodes, the *four-node roots*.

Theorem 5.3. *Assume Hypothesis 3.2 for $\widehat{X}_9$ and F . Let $x = (\phi, b, 0)$ be a root point with $\phi \in F$ and b a smooth point of N . Then, as germs of stratified spaces,*

$$\mathcal{V}_x \cong (\mathcal{K}_\phi \times N) \cup (F_\phi \times N \times \mathbb{C}^2/\mathbb{Z}_4), \quad (5.2)$$

glued along $F_\phi \times N \times \{0\}$, with F_ϕ of dimension $2 = h^{1,1}(X_{9,t}) - 1$. The hypermultiplet factor $N \times (\mathbb{C}^2/\mathbb{Z}_4 \setminus \{0\})$ of the Higgs branch off its root is a smooth quaternionic Kähler manifold of quaternionic dimension $124 = h^{2,1}(X_{9,t}) + 1$. There is no mixed branch.

Proof. We check (S1)–(S3) with $k = 4$. By Proposition 2.1(1) the curves of zero area on F are the four disjoint $(-1, -1)$ -curves C_1, C_2, e_1, e_2 . Their classes span a lattice of rank three with the single relation (2.5), so $a = (1, 1, -1, 1)$, and $\text{ann}(C_1, C_2, e_1, e_2)$ has dimension $6 - 3 = 3 = \dim F$, so F is open in it. The charge lattice is saturated, since the charge matrix (A.5) has Smith invariants $(1, 1, 1)$. The kinetic matrix is positive definite on F by Proposition 2.1(3). Theorem 5.2 applies with $m = 4$ and $h^{2,1}(\widehat{X}_9) = 122$. $\square$

Lemma 5.4. *In the situation of Theorem 5.3, the germ at x of the metric completion of $\mathcal{H}^{\text{reg}}$ is $F_\phi \times (N \times \mathbb{C}^2/\mathbb{Z}_4)$, and it coincides with the closure of $\mathcal{H}^{\text{reg}}$ in $\mathcal{V}_x$.*

Proof. Near x , $\mathcal{H}^{\text{reg}} = F_\phi \times (\mu^{-1}(0) \setminus N)/T^3$, and by clause (3) of Hypothesis 3.2 its hypermultiplet metric is bi-Lipschitz equivalent to the quotient metric. The link of the flat model is connected, so by Lemma 4.6 the completion of $(\mu^{-1}(0) \setminus N)/T^3$ near the root is $(\mu^{-1}(0) \cap U)/T^3$, which by Theorem 4.5 is the germ of $N \times \mathbb{C}^2/\mathbb{Z}_4$. Bi-Lipschitz equivalent

metrics have homeomorphic completions, and the completion of a product is the product of the completions. The neighbourhood U may be taken relatively compact and intrinsic distances dominate ambient ones, so the completion maps homeomorphically onto the closure of $\mathcal{H}^{\text{reg}}$ in $\mathcal{V}_x$. $\square$

Proposition 5.5. *In the situation of Theorem 5.3, let R_0 be the leading curvature of the transverse slice $\mathcal{Y}_p$ at the root, whose germ is $\mathbb{C}^2/\mathbb{Z}_4$ (Section 4.4). Then*

$$R_0 = 4\nu R^{(1)} + \mathcal{W}, \quad (5.3)$$

with $R^{(1)}$ as in Proposition 4.12 and $\mathcal{W} = c_\ell^{-4} \ell^*(R_{\text{hk}}|_{\ell(\mathbb{H})})$ ($c_\ell^2 = 4$) a $\mathbb{Z}_4$ -invariant anti-self-dual Weyl tensor on $\mathbb{C}^2$, where R_{hk} is the hyperkähler part of the curvature of M at the root, whose restriction to $V = \mathbb{H}^4$ is T^3 -invariant. The $\mathbb{Z}_4$ -invariant anti-self-dual Weyl tensors form a real space of dimension 3, and every element arises as $\mathcal{W}$ from some T^3 -invariant algebraic curvature tensor of hyperkähler type on V .

Proof. The first term is Proposition 4.12, and $\mathcal{W}$ is the restriction of R_{hk} to the quaternionic line $\ell(\mathbb{H})$. Algebraic curvature tensors of hyperkähler type on $\mathbb{H}^n$ correspond to quartic polynomials in the complex coordinates, compatibly with the real structure given by I_2 [46, 48]; restriction to $\ell(\mathbb{H})$ corresponds to pulling back the quartic by the complex-linear map underlying ℓ , and invariance under T^3 to invariance of the quartic. The T^3 -invariant quartics in $(z_i, \bar{z}_i)$ are spanned by the ten products $u_i u_j$ and the two generators X, Y of (A.6). With $\ell(\alpha, \beta)$ as in Section 4.4,

$$u_i \mapsto \pm \alpha \beta, \quad X \mapsto -\alpha^4, \quad Y \mapsto \beta^4, \quad u_i u_j \mapsto \pm \alpha^2 \beta^2,$$

so the image is spanned by α^4 , β^4 and $\alpha^2 \beta^2$. These are exactly the quartics invariant under the generator $(\alpha, \beta) \mapsto (i\alpha, -i\beta)$ of the stabiliser $\mathbb{Z}_4$ of $\ell(\mathbb{H})$, which acts on the monomials $\alpha^{4-l} \beta^l$ by i^{4-2l} . The real structure induced on the quartics in (α, β) by the quaternionic structure I_2 exchanges α^4 and β^4 up to conjugation and fixes $\alpha^2 \beta^2$, so the real invariant tensors are $c \alpha^4 + \bar{c} \beta^4 + t \alpha^2 \beta^2$, $c \in \mathbb{C}$, $t \in \mathbb{R}$, a space of real dimension three. $\square$

The three directions have a direct interpretation. The products $u_i u_j$, invariant under the full torus $U(1)^4$, give $\alpha^2 \beta^2$, the direction invariant under the triholomorphic circle $(\alpha, \beta) \mapsto (e^{i\theta} \alpha, e^{-i\theta} \beta)$; it is the Taub–NUT direction, spanned by the curvature at the nut of the Taub–NUT space, whose Gibbons–Hawking potential [49] has a nonzero constant term (checked in Appendix A). The invariants X and Y of Lemma 5.1 have degree m ; because $m = 4$ here they are quartic and give α^4 and β^4 , the two directions that break this circle. For a general face with one relation, $\Gamma = \mathbb{Z}_m$ acts on $\ell(\mathbb{H})$ by $(\alpha, \beta) \mapsto (\zeta \alpha, \zeta^{-1} \beta)$ and hence on the quartics $\alpha^{4-l} \beta^l$ by ζ^{4-2l} , so the Weyl part lies in a real space of dimension 5 for $m = 2$, 3 for $m = 4$ and 1 for all other m .

The proof uses only the quaternionic Kähler manifold M at the root and the charges, so the same statement holds in the setting of Theorem 6.2, and for the metric of the type IIA

hypermultiplet moduli space $\mathcal{M}_H(X_t)$ when the identification (W-IIA) of Section 6.4 is an isometry. In that sense Proposition 5.5 makes the “non-universal” correction of [10, Sect. 6] precise at leading order: a term fixed by supergravity and a term in a three-dimensional space (a number specific to $m = 4$), on which T^3 -invariance imposes no further linear constraint. Whether every such $\mathcal{W}$ arises from a quaternionic Kähler manifold with this isometric torus action is not addressed.

5.3 The del Pezzo facet

Through the smoothing of Proposition 2.2, $H^2(X_{9,t}; \mathbb{R})$ is identified with $\text{ann}(C_1, C_2, e_1, e_2)$, spanned by H_2, H_3, H_4 , with the same cubic form (A.1); we use the coordinates y_2, y_3, y_4 on both. The facet $F_0 = \{y_2 = 0\}$ of $\bar{F}$ is the locus where the whole surface E collapses; there the light sector is the rank-one E_2 theory together with the two nodes [3, 4]. It is not a boundary of the Higgs branch. By [4, eq. (5.25)] the Mori cone of $X_{9,t}$ lies in the cone spanned by the degrees $(1, 1, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ in the basis (H_2, H_3, H_4) , so

$$\mathcal{K}(X_{9,t}) \supseteq \{y_2 + y_3 > 0, y_3 > 0, y_4 > 0\}, \quad (5.4)$$

which contains the relative interior of F_0 . The regular Higgs branch therefore continues across the facet.

Theorem 5.6. *Assume Hypothesis 3.2 for $\hat{X}_9$ and F . Let $y_0 = y_2 H_2 + y_3 H_3 + y_4 H_4$ lie in the region (5.4), normalised by $J^3 = 6$, and let x be the limit point of the regular Higgs vacua as the charged hypermultiplet expectation values, equivalently the smoothing parameter t , tend to zero, with the vector moduli fixed at y_0 and the neutral hypermultiplets fixed at a smooth point $b \in N$. Then the germ at x of the metric completion of $\mathcal{H}^{\text{reg}}$ is*

$$(\text{germ of } \mathcal{K}(X_{9,t}) \cap \{J^3 = 6\} \text{ at } y_0) \times (N \times \mathbb{C}^2/\mathbb{Z}_4), \quad (5.5)$$

whose vector factor is a full two-dimensional neighbourhood of y_0 . In particular this holds at every point y_0 of the relative interior of the del Pezzo facet F_0 .

Proof. By Lemma 3.3, near x the regular Higgs branch is the metric product of the vector germ and a fixed manifold $\mathcal{M}_H^{\text{reg}}$, and the metric completion of such a product is the product of the completions. At a point of F , Lemma 5.4 gives the completion as $F_\phi \times (N \times \mathbb{C}^2/\mathbb{Z}_4)$, so the completion of $\mathcal{M}_H^{\text{reg}}$ at the root is $N \times \mathbb{C}^2/\mathbb{Z}_4$. Since $\mathcal{M}_H^{\text{reg}}$ does not depend on the vector moduli, the same holds over y_0 . $\square$

Theorem 5.6 says nothing about the Coulomb branch at x , which at F_0 contains the interacting E_2 point, and it describes the completion of the regular Higgs branch, not the Higgs branch of the interacting theory at the root. The roots over the rest of the region (5.4), and the boundary $y_2 = -y_3$, beyond which Theorem 5.6 does not apply, are described in Section 7.2.

6 Four dimensions: type IIA string theory

The arguments of Sections 4 and 5 use the spacetime dimension only through the supersymmetry conditions (3.3). In this section we impose the corresponding conditions of four-dimensional $N = 2$ supergravity directly. The resulting statement concerns type IIA string theory on $\widehat{X}$ and, through its hypermultiplet moduli space, the setting of Greene, Morrison and Vafa [10, Sect. 6]. Type IIA on $\widehat{X}$ is M-theory on $\widehat{X} \times S^1$, but the argument does not pass through the five-dimensional theory: it is carried out directly in four dimensions, where the massless states at the root carry no D0-brane charge, so that no tower of Kaluza–Klein (D0) states has to be truncated.

6.1 Supersymmetric Minkowski vacua in four dimensions

We use the conventions of [5, 50]: n_V vector multiplets with special Kähler manifold $\mathcal{S}$, homogeneous coordinates X^Λ , $\Lambda = 0, \dots, n_V$, a prepotential $\mathcal{F}(X)$, symplectic sections $(\mathcal{L}^\Lambda, \mathcal{M}_\Lambda) = e^{K/2}(X^\Lambda, \partial_\Lambda \mathcal{F})$, $f_i^\Lambda = \nabla_i \mathcal{L}^\Lambda$ and metric $g_{i\bar{j}}$; hypermultiplets with quaternionic Kähler manifold M of reduced scalar curvature $\nu < 0$ fixed by supergravity; and an electric gauging of a torus acting on M by isometries, with Killing vectors k_Λ and triplets of moment maps $\vec{P}_\Lambda$, and acting trivially on $\mathcal{S}$. For a light state of charge $q \in \mathbb{Z}^{n_V+1}$ we write $Z_q = e^{K/2} q_\Lambda X^\Lambda$ for its central charge.

For configurations with constant scalars and vanishing field strengths the gravitino, gaugino and hyperino variations vanish if and only if $\vec{P}_\Lambda \mathcal{L}^\Lambda = 0$, $g^{i\bar{j}} \bar{f}_{\bar{j}}^\Lambda \vec{P}_\Lambda = 0$ and $k_\Lambda \mathcal{L}^\Lambda = 0$ [5, Sects. 2.1–2.3, 2.4.2]. Since the $\vec{P}_\Lambda$ are real and, where $g_{i\bar{j}}$ is positive definite, the $(n_V + 1) \times (n_V + 1)$ matrix $(\bar{\mathcal{L}}^\Lambda, f_i^\Lambda)$ is invertible [51, Lemma B.5, Cor. B.6], the conditions become

$$\vec{P}_\Lambda = 0 \quad \text{for all } \Lambda, \quad k_\Lambda \mathcal{L}^\Lambda = 0, \quad (6.1)$$

as stated in [5, Sect. 2.4.2]. Positive definiteness of $g_{i\bar{j}}$ is equivalent to negative definiteness of the imaginary part of the gauge kinetic matrix $\mathcal{N}_{\Lambda\Sigma}$ [51, eq. (4.45)]; it takes the place of condition (S3). We write n_N for the quaternionic dimension of the manifold N of neutral hypermultiplets. Given charge vectors $q_1, \dots, q_k$ of k charged hypermultiplets, a *root point* is a point $x = (z_0, b, 0)$ with $z_0 \in \mathcal{S}$ at which $Z_{q_1}, \dots, Z_{q_k}$ vanish, $b \in N$, and vanishing charged hypermultiplets; $\mathcal{V}_x$ is defined as in Definition 3.1, with z_0 in place of ϕ and the theory of the following hypothesis.

Hypothesis 6.1 (regularity, four dimensions). Near a root point $x = (z_0, b, 0)$ the dynamics below the lightest massive state is a four-dimensional $N = 2$ gauged supergravity as described above, in which exactly k charged hypermultiplets are light, with: $g_{i\bar{j}}$ positive definite at z_0 ; a smooth quaternionic Kähler manifold M of quaternionic dimension $n_N + k$ containing the manifold N of neutral hypermultiplets, including the corrections to its metric from D2-instantons on three-cycles that do not vanish at the root and from NS5-instantons; charge vectors $q_1, \dots, q_k$ of the light states spanning a lattice of rank $k - 1$ with the single relation $\sum_i a_i q_i = 0$, a primitive and every $a_i \neq 0$; no neutral hypermultiplet charged; and b a smooth

point of N . Moreover, the Coulomb branch through x consists of supersymmetric Minkowski vacua, and the higher-derivative couplings do not change the set of such vacua.

As in five dimensions (Section 7.5), D2-instantons on three-cycles that do not vanish and NS5-instantons correct the metric of M smoothly and preserve the gauged isometries. We expect, as in the three-dimensional analysis of [7], that the Euclidean D2-branes on the vanishing three-spheres, which produce the singularity of the neutral metric in [6], describe the same singular point in terms of the neutral fields alone: in the Wilsonian description their effect is carried by the light charged hypermultiplets and, through the massive D2–D0 states, by smooth corrections to M . This is part of the content of the identification (W-IIA) of Section 6.4.

Theorem 6.2. *Assume Hypothesis 6.1, write $Z_i = Z_{q_i}$, and let $\mathcal{S}_0 = \{Z_1 = \cdots = Z_k = 0\}$. Then $\mathcal{S}_0$ is a complex submanifold of $\mathcal{S}$ of codimension $k - 1$ near z_0 and, writing $(\mathcal{S}, z_0)$ and $(\mathcal{S}_0, z_0)$ for germs, as germs of stratified spaces*

$$\mathcal{V}_x \cong ((\mathcal{S}, z_0) \times N) \cup ((\mathcal{S}_0, z_0) \times N \times \mathbb{C}^2/\mathbb{Z}_m), \quad m = \sum_i |a_i|, \quad (6.2)$$

glued along $(\mathcal{S}_0, z_0) \times N \times \{0\}$. There is no mixed branch; the hypermultiplet factor of the Higgs branch off its root is a smooth quaternionic Kähler manifold of quaternionic dimension $n_N + 1$, and at distance r from the root, measured in the metric of M normalised as in [5] ($\kappa^2 = 1$), its metric is the flat orbifold metric of $\mathbb{C}^2/\mathbb{Z}_m$ times the flat metric on $T_b N$ up to relative corrections of order r^2 .

Proof. In a frame with a prepotential the matrix $(X^\Lambda, \partial_j X^\Lambda)$ is invertible [51, Sect. 4, on the existence of a prepotential], so a combination $\sum_i c^i q_i$ annihilating both X and ∂X at a point of $\{q \cdot X = 0\}$ vanishes; since $\sum_i a_i Z_i \equiv 0$ and the q_i have rank $k - 1$, $\mathcal{S}_0$ has codimension $k - 1$. On the Coulomb branch through x the conditions (6.1) hold with the charged hypermultiplets set to zero, so the moment map vanishes at the point $p \in M$ over b , and on N near b by Lemma 4.3. The torus acting effectively on the charged hypermultiplets is the image of $U(1)^{n_V+1}$ in $U(1)^k$, whose cocharacter lattice is $a^\perp \cap \mathbb{Z}^k$, so Lemma 5.1 applies without a saturation assumption. Section 4 and Lemma 5.1 do not refer to the spacetime dimension, and Theorem 4.5, Proposition 4.7 and Lemma 5.1 give the hypermultiplet germ $N \times \mathbb{C}^2/\mathbb{Z}_m$ and the metric statement as in the proof of Theorem 5.2. For the vector conditions, $k_\Lambda \mathcal{L}^\Lambda$ is the fundamental field of the element ξ of $\text{Lie}(T) \otimes \mathbb{C}$ determined by $(Z_1, \dots, Z_k)$. Off the root the effective torus acts freely near p , so the real and imaginary parts of ξ vanish, that is $Z_i = 0$ for all i ; conversely $Z = 0$ gives $k_\Lambda \mathcal{L}^\Lambda = 0$. On the Coulomb branch every z is allowed. The absence of a mixed branch follows as in Theorem 5.2. $\square$

The Higgs branch has vector factor $(\mathcal{S}_0, z_0)$ of complex dimension $n_V - (k - 1)$. For type IIA on $\widehat{X}$ at large volume (Section 6.2), with the D0-charges fixed by the gauge choice $B \cdot C_i = 0$ made there, the root points with $J \in F$ are the points of $\mathcal{E}_F \times N$, and $n_V - (k - 1) = h^{1,1}(X_t)$, the number of vector multiplets of type IIA on the smoothing.

6.2 Type IIA string theory at the massless locus

Let $\widehat{X}$ and F satisfy (S1)–(S3). We use the large-volume symplectic frame of type IIA on $\widehat{X}$, with complexified Kähler moduli $w = B + iJ$, where $w^A = X^A/X^0$ and the prepotential is $\mathcal{F} = -\frac{1}{6}\kappa_{ABC}X^AX^BX^C/X^0$ plus worldsheet-instanton terms. In this frame a D2-brane on a class β bound to n D0-branes has holomorphic central charge $\beta \cdot w - n$, up to the factor $e^{K/2}$ and the sign conventions of [52, eq. (A.10)]; D2- and D0-charges are electric [52]. The vector multiplets gauge no isometry of $\mathcal{S}$. We call

$$\mathcal{E}_F = \{w = B + iJ : J \in F, B \cdot C_i \in \mathbb{Z} \text{ for all } i\}$$

the *massless locus* over F , the locus of the complexified Kähler cone at which D2-branes on the curves C_i become massless. The large gauge transformations $B \mapsto B + D$, $D \in H^2(\widehat{X}; \mathbb{Z})$, identify its components and shift the D0-charges; since the charge lattice is saturated by (S2), we can use them to set $B \cdot C_i = 0$. A D2-brane on a class in the span of the C_i then has central charge proportional to n , so among these states only the D2-branes on the C_i with $n = 0$ are massless; by (S1) and the vanishing of multiple-cover and higher-genus invariants of isolated $(-1, -1)$ -curves [24, 25] they give k hypermultiplets with charges $q_{iA} = H_A \cdot C_i$, $q_{i0} = 0$. D4- and D6-brane central charges receive worldsheet-instanton corrections and can vanish at finite volume, for instance for D4-branes on a small divisor near the del Pezzo facet of X_9 . We therefore work at large volume, $J = \rho J_0$ with J_0 in a compact subset of F and $\rho \geq \rho_0(J_0)$, and assume, as part of Hypothesis 6.1, that there the central charges of all other BPS states are bounded away from zero (for D0-branes and D2–D0 states on the span of the C_i this holds exactly, $|n| \geq 1$); $\rho_0(J_0) \rightarrow \infty$ as J_0 approaches the boundary of F .

Proposition 6.3. *For the cubic prepotential, with $L(J)_{AB} = \kappa_{ABC}J^C$, $v = L(J)J$ and $\text{vol} = \frac{1}{6}\kappa(J, J, J)$, one has $e^{-K} = \frac{4}{3}\kappa(J, J, J)$ and the special Kähler metric on the complexified Kähler cone is*

$$g_{A\bar{B}} = \frac{1}{4\text{vol}} \mathcal{G}_{AB}, \quad \mathcal{G} = -L(J) + \frac{vv^\top}{4\text{vol}}, \quad (6.3)$$

where $\mathcal{G}$ is the Hodge metric on harmonic two-forms of (S3), that is the five-dimensional gauge kinetic matrix [4, eqs. (4.1), (4.6)]. In particular, on F the metric of the cubic prepotential is positive definite if and only if (S3) holds.

Proof. With $X^0 = 1$, $e^{-K} = i(\bar{X}^\Lambda \partial_\Lambda \mathcal{F} - X^\Lambda \bar{\partial}_\Lambda \bar{\mathcal{F}}) = \frac{4}{3}\kappa(J, J, J)$. Since K depends on J only, $\partial_{w^A} \partial_{\bar{w}^B} K = \frac{1}{4} \partial_{J^A} \partial_{J^B} K$, and with $\partial_A \kappa(J, J, J) = 3v_A$, $\partial_A \partial_B \kappa(J, J, J) = 6L_{AB}$ one obtains $g = \frac{1}{4}(-6L/\kappa(J, J, J) + 9vv^\top/\kappa(J, J, J)^2)$, which is (6.3) with $\kappa(J, J, J) = 6\text{vol}$. $\square$

The corrections to the Wilsonian prepotential near $\mathcal{E}_F$ are the constant $\zeta(3)\chi$ term, of relative order ρ^{-3} ; worldsheet instantons on classes of positive area, exponentially small; and the regular part of $\text{Li}_3(e^{2\pi i w \cdot C_i})$ at the shrinking curves, whose non-analytic part, proportional to $(w \cdot C_i)^2 \log(w \cdot C_i)$, is the one-loop contribution of the light hypermultiplets and is absent from the Wilsonian prepotential. The regular part changes g by a relative amount of order ρ^{-1} in the directions normal to $\mathcal{S}_0$, with a coefficient fixed by the Wilsonian subtraction scale;

for any fixed subtraction scale, the positive definiteness in Hypothesis 6.1 follows from (S3) for ρ large. Proposition 2.1(3) establishes (S3) for X_9 .

6.3 The vanishing three-spheres

Lemma 6.4. *Let X_t be a smoothing of the contraction X_F of Section 3.1, and $S_i \subset X_t$ the vanishing three-sphere of the i -th node. For suitable orientations of the S_i , $[S_i] = a_i[S]$ in $H_3(X_t; \mathbb{Z})$ for a class $[S]$ generating the span of the $[S_i]$. If the relation $\sum_i a_i[C_i] = 0$ holds in $H_2(\hat{X}; \mathbb{Z})$ and not only modulo torsion, then $[S]$ is primitive.*

Proof. The complements $\hat{X} \setminus \bigcup_i C_i$ and $X_t \setminus \bigcup_i S_i$ are diffeomorphic, and the linking three-sphere of C_i is identified with S_i . In the exact sequence $H_4(\hat{X}) \rightarrow \bigoplus_i H_4(\hat{X}, \hat{X} \setminus C_i) \cong \mathbb{Z}^k \rightarrow H_3(\hat{X} \setminus \bigcup C_i)$ the first map is $D \mapsto (D \cdot C_i)_i$, whose image is $a^\perp \cap \mathbb{Z}^k$ by (S2), and the second sends the i -th generator to the linking sphere. The map $H_3(X_t \setminus \bigcup S_i) \rightarrow H_3(X_t)$ is injective, because $H_4(X_t, X_t \setminus \bigcup S_i) \cong H^2(\bigcup S_i) = 0$ by Lefschetz duality. So the span of the $[S_i]$ is $\mathbb{Z}^k / (a^\perp \cap \mathbb{Z}^k) \cong \mathbb{Z}$, via $c \mapsto a \cdot c$. For primitivity consider $H_3(X_t) \rightarrow H_3(X_t, X_t \setminus \bigcup S_i) \cong \mathbb{Z}^k \rightarrow H_2(X_t \setminus \bigcup S_i) \cong H_2(\hat{X} \setminus \bigcup C_i) \cong H_2(\hat{X})$, the last isomorphism because $H_j(\hat{X}, \hat{X} \setminus \bigcup C_i) \cong H^{6-j}(\bigcup C_i) = 0$ for $j = 2, 3$. The first map is $\gamma \mapsto (\gamma \cdot S_i)_i$ and the second sends the i -th generator to $\pm[C_i]$, so the image of the first is the integral relation lattice of the curves, $\mathbb{Z}a$. Hence some γ has $\gamma \cdot S = \pm 1$, and $[S]$ is primitive. $\square$

For unit coefficients the k vanishing spheres are homologous up to orientation: this is the configuration of k homologous vanishing three-cycles of [6, 10]. For X_9 , Lemma 6.4 with (2.5) gives four such spheres.

6.4 The expectation of Greene, Morrison and Vafa

Greene, Morrison and Vafa argued from the rigid field theory that near homologous vanishing three-cycles the type IIA hypermultiplet moduli space has, with gravity decoupled, an A_{k-1} singularity, k the number of cycles, and wrote: “When we turn on gravity we do expect that the qualitative features we have found will persist but the actual metric will get corrected in a non-universal way, depending on how the vanishing cycle sits in the Calabi–Yau, and the total space will be a quaternionic manifold” [10, Sect. 6]. Here “gravity decoupled” is the limit in which the four-dimensional dilaton, a coordinate of the universal hypermultiplet, is sent to weak coupling while approaching the conifold locus, so that the relevant piece of the quaternionic Kähler metric scales to a hyperkähler one [10, Sect. 6]; “finite Planck mass” means the quaternionic Kähler metric at fixed ν .

Under (W-IIA) below, we identify the neutral manifold N of $\hat{X}$ with the locus in the hypermultiplet moduli space of the smoothings X_t where they degenerate to X_F : its points are the complex structures of X_F with their Ramond–Ramond axions and the scalars of the universal hypermultiplet. We call $N \times \{0\}$ the *root locus*; it lies in the conifold locus of $\mathcal{M}_H(X_t)$, where the vanishing cycles have zero period. To compare the vacuum space of Theorem 6.2 with the moduli space of type IIA on X_t we assume, at large volume:

(W-IIA) near x , the Higgs vacua off the root are type IIA compactifications on the smoothings X_t , and the hypermultiplet moduli space $\mathcal{M}_H^{\text{reg}}$ of these (the same manifold as in Section 3.3, by [36, Sect. 2]) is bi-Lipschitz equivalent near the root to $(\mu^{-1}(0) \setminus N)/T$ with the quotient metric of [1].

This is the four-dimensional counterpart of clause (3) of Hypothesis 3.2, and it is the assumption on which the comparison with [10] rests. Without gravity its analogue holds for the known model metrics: the hyperkähler metrics obtained from Euclidean D2-branes on k homologous vanishing spheres [6, 7] (constructed rigorously for $k = 1$ in [9], whose Remarks 3.14 and 4.44 indicate the extension to multiplicity k) have the same A_{k-1} point as the hyperkähler quotient of the k light hypermultiplets. A proof at finite Planck mass would require control of the instanton-corrected quaternionic Kähler metric near the conifold locus. The twistorial construction [53] describes the D-instanton corrections to this metric, NS5-instantons aside, through the invariants of the wrapped branes, but its behaviour where the central charge of a wrapped brane vanishes has not been analysed there, and it does not settle (W-IIA).

Corollary 6.5. *Let $\widehat{X}$ and F satisfy (S1)–(S3), and assume Hypothesis 6.1 and (W-IIA) for type IIA on $\widehat{X}$ at large volume. At a point of the root locus the germ of the metric completion of the hypermultiplet moduli space of X_t is homeomorphic to $N \times \mathbb{C}^2/\mathbb{Z}_m$ with $m = \sum_i |a_i|$, and it is a smooth quaternionic Kähler manifold off $N \times \{0\}$. If the equivalence in (W-IIA) is an isometry, the metric is the flat orbifold metric up to relative corrections of order r^2 .*

Proof. The argument of Lemma 5.4 for the hypermultiplet factor, with Theorem 6.2 in place of Theorem 5.2 and (W-IIA) in place of clause (3) of Hypothesis 3.2; the metric clause follows from Proposition 4.7 when the equivalence is an isometry. $\square$

For unit coefficients $m = k$, the number of homologous vanishing spheres. Near the root the rest of the conifold locus consists of smooth points of the completion: in $\mathbb{C}[X, Y, s]/(XY - cs^m)$, $c \neq 0$ as in Lemma 5.1, it is $\{s = 0, (X, Y) \neq 0\}$. For coefficients $|a_i| \geq 2$ the rigid instanton picture is consistent with this description of the root. Near the conifold locus the rigid metric has Gibbons–Hawking form, with coordinates the period $\int_S \Omega$ and the Ramond–Ramond phase $\theta = \int_S C_3$ and a harmonic potential. A BPS state of charge n with respect to $[S]$ (for the sphere S_i , $n = |a_i|$) in the three-dimensional description obtained by compactification on a circle [7] contributes n singular points to this potential, at $\theta \in 2\pi\mathbb{Z}/n$, each of which is an A_{n-1} point [54, Sect. 4.1], and the one-loop potential is a sum over the light hypermultiplets [7, 54]. Hence at $\theta = 0$, where the centres of all the S_i coincide, the singularity is of type A_{m-1} with $m = \sum_i |a_i|$. The rigid picture also places orbifold points at other values of θ , outside the germ; we make no statement about them.

Higgs-side region	Coulomb-side root	light states at the root
$y_2 > 0$	$J = y \in F$	four free hypermultiplets
$y_2 = 0$	$J = y \in F_0$	E_2 theory and two nodes
$\max(-y_3, -2y_4) < y_2 < 0$	$J = y + \frac{y_2}{2}E$ (after a flop)	E_1 theory and two nodes
$y_2 = -2y_4 > -y_3$	wall	$D_2 \cup C_1 \cup C_2$ contracts to a point
$-y_3 < y_2 < -2y_4$	$J = y + \frac{y_2}{2}E + \frac{y_2+2y_4}{2}D_2$	two collapsed surfaces

Table 1. Roots of the Higgs branch over the region $y_3, y_4 > 0$ of the Kähler cone of $X_{9,t}$, with $y = y_2H_2 + y_3H_3 + y_4H_4$. At $y_2 = -y_3$ a plane in $X_{9,t}$ collapses and the vector-multiplet metric on the Higgs side degenerates.

7 Comparison and discussion

7.1 The rigid description

In the rigid model the transition of X_9 has two descriptions. At the four-node roots it is the hyperkähler quotient of four hypermultiplets by a rank-three torus. At the E_2 point it is built from the two nodal hypermultiplets and the E_2 theory, whose reduced Higgs branch is the cone $\mathbb{C}[J_+, J_0, J_-]/(J_+J_- - J_0^2) = \mathbb{C}^2/\mathbb{Z}_2$ generated by the moment maps $J_\pm, J_0$ of its $SU(2)$ flavour symmetry; the two bulk vector multiplets gauge a rank-two torus acting on the nodes and on this cone [4, eqs. (2.3), (3.10)]. Both give $\mathbb{C}[X, Y, s]/(XY - s^4)$, the sign of s^4 being a convention (compare (A.6)). The two presentations are related by reduction in stages: quotienting the four-hypermultiplet model first by the third row of (A.5) produces the $\mathbb{C}^2/\mathbb{Z}_2$ cone of the E_2 theory, on which the first two rows act as the rank-two gauging. Theorems 5.3 and 5.6 show that this rigid germ survives at finite Planck mass under Hypothesis 3.2, at the four-node roots and, for the completion of the regular Higgs branch, at the del Pezzo facet.

7.2 The other roots of X_9

Theorem 5.6 applies over the whole region (5.4), but the root over which the regular Higgs vacua end changes across the region. Table 1 lists the roots over $y_3, y_4 > 0$. The regular Higgs vacua over $y_2 > 0$ end at the four-node roots. Those over $\max(-y_3, -2y_4) < y_2 < 0$ end at the locus where the image of E in the flop of $\widehat{X}_9$ along the curve $h - e_1 - e_2$, a copy of $\mathbb{P}^1 \times \mathbb{P}^1$, collapses to a point, giving the rank-one E_1 theory together with the two nodes. Across the wall $y_2 = -2y_4$ the toric divisor D_2 of Section 2.1 collapses together with C_1 and C_2 , and beyond it two surfaces are collapsed at the root. The cubic prepotential is continuous across each of these walls, as an identity among intersection numbers. The lists of curves of zero area rest on the Gopakumar–Vafa invariants of all classes of total degree at most eight; for curves $c = dh - d_1e_1 - d_2e_2$ inside E , whose area at the E_1 root is $\frac{y_2}{2}(d_1 + d_2 - d)$, they hold in all degrees.

These rows are not used in the proofs. At all these roots Theorem 5.6 describes the completion of the regular Higgs branch; whether further Higgs-type components meet the Coulomb branch there has not been examined. Theorem 5.6 does not apply beyond $y_2 = -y_3$.

There a divisor isomorphic to $\mathbb{P}^2$ in $X_{9,t}$ collapses, giving the E_0 theory on the smooth side [4], the Hessian

$$\det \text{Hess}(\kappa|_{X_{9,t}}) \propto (y_2 + y_3)(y_2 + y_3 + 3y_4)(y_2 + 2y_3 + 4y_4) \quad (7.1)$$

of the cubic form of $X_{9,t}$ vanishes, and the regular Higgs branch itself degenerates.

The facet $y_3 = 0$ of $\overline{F}$, which we call the $SU(2)$ facet because the curve class β_3 becomes massless there and gives an $SU(2)$ vector multiplet, provides a consistency test of the definition of the Higgs branch as a metric completion, at a root with a Lagrangian description that lies outside the scope of Theorem 4.5. If no class of degree $(0, j, 0)$, $j \geq 1$, is effective on $X_{9,t}$ (for $j = 1, 2$ the Gopakumar–Vafa invariants in these degrees sum to zero), the relative interior of this facet is interior to $\mathcal{K}(X_{9,t})$, and the completion there again has hypermultiplet germ $N \times \mathbb{C}^2/\mathbb{Z}_4$. Among classes of total degree at most eight, the light states on the Coulomb side of this facet are an $SU(2)$ vector multiplet from the class β_3 , two doublets $(e_i, e_i + \beta_3)$ and the singlets C_1, C_2 ; the $SU(2)$ coupling is finite, since it is set by the area $y_2 > 0$ of $h - e_1 - e_2$. The rigid model of this $SU(2) \times U(1)^3$ theory is again exactly $\mathbb{C}^2/\mathbb{Z}_4$. The level set of the complex moment map has a single component of stable points, on which the two doublets are aligned, and the invariant ring of the whole level set is $\mathbb{C}[A, B, s]/(AB + s^4)$. This agrees with the germ of the completion, but it concerns the rigid model only: the generic stabiliser there is one-dimensional, so Theorem 4.5 does not apply at finite Planck mass.

7.3 Instanton geometry without gravity

In type IIA string theory on a threefold near a conifold degeneration, the hypermultiplet moduli space receives corrections from Euclidean D2-branes on the vanishing three-sphere, and the corrected geometry near k homologous vanishing three-cycles has a $\mathbb{C}^2/\mathbb{Z}_k$ singularity [6, 7]; the multiplicity is the Gopakumar–Vafa invariant of the vanishing class [8], and the smooth model geometries of this kind were constructed rigorously for $k = 1$ in [9], where the extension to multiplicity k is indicated [9, Rems. 3.14, 4.44]. For X_9 the relation (2.5) has unit coefficients, so the four vanishing three-spheres of the smoothing of X_F lie in the classes $\pm[S]$ of one primitive class, and this picture gives an A_3 point, in agreement with Theorem 5.3. These analyses rest on rigid, gravity-decoupled arguments. Greene, Morrison and Vafa derived $\mathbb{C}^2/\mathbb{Z}_k$ for k hypermultiplets with one relation of this kind [10, Sect. 3] and expected that with gravity the qualitative features of the four-dimensional hypermultiplet space persist while the metric is corrected [10, Sect. 6]. Theorem 5.2 and Proposition 4.7 make the five-dimensional counterpart precise, under the regularity hypothesis, for every face satisfying (S1)–(S3), X_9 included: the germ is unchanged and the metric changes only at relative order r^2 . Corollary 6.5 gives the four-dimensional statement about the type IIA hypermultiplet moduli space, for the smoothings of the contractions X_F of faces satisfying (S1)–(S3) and under the identification (W-IIA), and Proposition 5.5 identifies the leading correction to the Wilsonian quotient metric, which is the moduli-space metric if (W-IIA) holds isometrically. Theorem 5.2 does not cover faces whose curves satisfy several relations; there

proper subsets of the light states can condense, the flat quotient has non-isolated singularities, and Theorem 4.5 would need a stratified extension.

7.4 A census of one-relation faces

Theorem 5.2 applies to every face with one relation, and X_9 is one example. Its proof uses the light spectrum, the openness of F in the annihilator of the classes of the light states, and the positivity of the kinetic matrix. The census below tests these three inputs and, where toric data suffice, the geometric part of (S1), namely that the light states come from disjoint $(-1, -1)$ -curves; it does not test the saturation in (S2), which only normalises the torus. To see how often these hold, and whether coefficients $|a_i| \geq 2$ occur, we examined all favourable reflexive polytopes (those for which every divisor class of the hypersurface is the restriction of an ambient toric divisor class) in four dimensions with $2 \leq h^{1,1} \leq 5$ in the classification of Kreuzer and Skarke [55], and a random sample of 2000 of the 17101 polytopes with $h^{1,1} = 6$, of which the favourable ones were examined. For each polytope we took all fine regular star triangulations when there are at most forty, and otherwise up to forty distinct ones obtained from random heights. For each triangulation we took the closed cone of Kähler classes of the hypersurface pulled back from the toric ambient space, which we call its toric Kähler cone, and every proper face F of this cone. At an interior point of F we listed the curve classes of zero area with nonzero genus-zero Gopakumar–Vafa invariant n_β , up to degree six with respect to a grading vector positive on the Mori cone of the toric ambient space, computed from the mirror periods [56] with the software of [57]. We call F a *one-relation face* when

- (i) all these invariants are positive;
- (ii) the volume and the volumes of all prime toric divisors are positive, computed exactly from the intersection numbers; together with (i) this excludes massless vector multiplets (invariant -2) and prime toric divisors that are ruled surfaces collapsing onto curves of genus $g \geq 2$, whose fibre class has the positive invariant $2g - 2$;
- (iii) the light spectrum, with n_β hypermultiplets of charge β and $k = \sum_\beta n_\beta$, has exactly one relation among its charges, with all coefficients nonzero (so $k \geq 2$);
- (iv) F is open in the annihilator of these classes, as in (S1);
- (v) the gauge kinetic matrix is positive definite at that point.

A face found from several triangulations of one polytope, with the same classes up to sign, was counted once. Table 2 gives the result.

Most one-relation faces are flop walls, $k = 2$, where two hypermultiplets of equal charge become massless, the situation of the gauged supergravities of [2]; faces with $k = 3, 4$ or 5 occur in 210 polytopes. In no case does a coefficient $|a_i| \geq 2$ occur. For a flop wall, on which all classes of zero area lie on one ray, this would require exactly one isolated curve in each of two classes $p\beta_0$ and $p'\beta_0$, $p < p'$, and none in the other multiples, for example β and 2β ;

$h^{1,1}$	scanned	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$ a_i \geq 2$
2	36	1 (1)	–	–	–	0
3	243	24 (21)	1 (1)	–	–	0
4	1185	335 (243)	11 (8)	–	–	0
5	4897	2895 (1727)	83 (56)	113 (63)	1 (1)	0
6	1948	1978 (946)	54 (33)	108 (52)	1 (1)	0

Table 2. One-relation faces, faces satisfying (i)–(v), by $h^{1,1}$ and the number $k = \sum_{\beta} n_{\beta}$ of light hypermultiplets, in Calabi–Yau hypersurfaces from favourable Kreuzer–Skarke polytopes (all polytopes with $h^{1,1} \leq 5$; the favourable ones among a random sample of 2000 of the polytopes with $h^{1,1} = 6$). “Scanned” is the number of favourable polytopes examined. Entries give the number of faces and, in parentheses, of polytopes; a polytope can appear in several columns. All coefficients satisfy $|a_i| = 1$, so $m = k$.

the relation is then $a = (p', -p)/\gcd(p, p')$. Here the length d of a flopping curve is that of Katz and Morrison [58], the coefficient of the curve in the fundamental cycle of a general hyperplane section through the singular point. A generic deformation splits a flopping curve of length d into $n_{j\beta}$ isolated curves in the classes $j\beta$, $j \leq d$ [59], and for the flopping curves of length two in [60] one has $(n_{\beta}, n_{2\beta}) = (5, 1)$, so this pattern does not arise from them. Whether faces with $m \neq k$ occur at all remains open; Theorem 5.2 covers them if they do.

Genus-zero invariants alone do not distinguish k disjoint $(-1, -1)$ -curves, as for X_9 , from other configurations with the same light spectrum, for instance two curves meeting in a point, contracted to a compound Du Val point of type cA_2 (for $k = 3$), or a single curve with $n_{\beta} = 2$ in place of two curves in one class. The geometric reading of Theorem 5.2 as a transition through a threefold with k nodes needs the first alternative. We tested it with toric curves. For a two-dimensional cone $\{i, j\}$ of the fan, $C_{ij} = D_i \cdot D_j \cdot X$ is, for generic X , empty or a disjoint union of ℓ torus-translates of a smooth curve of genus g , with $\ell > 1$ only if $g = 0$; its class is given by the intersection numbers $\kappa(D, D_i, D_j)$, and by adjunction $\kappa_{iij} + \kappa_{ijj} = \ell(2g - 2)$, so normal degrees $(\kappa_{iij}, \kappa_{ijj}) = (-1, -1)$ force a single $(-1, -1)$ -curve and $(-2, -2)$ two disjoint ones in half the class. Two such curves with one common divisor meet in κ_{ijl} points, and curves without a common divisor are disjoint, since X avoids the torus-fixed points. If every ray of the fan is a ray of the fan of the ambient toric morphism defined by F (equivalently, $J^3 \cdot D_i > 0$ on the ambient space for every prime toric divisor D_i), every curve of zero area is a component of some C_{ij} ; this condition depends on F alone. We say that toric data *decide* a face satisfying it if, in some triangulation whose toric Kähler cone contains F , the zero-area curves C_{ij} are exactly the light states, with normal degrees $(-1, -1)$ (for a light class with $n_{\beta} = 2$, either two such curves or one C_{ij} in twice the class with degrees $(-2, -2)$), and are pairwise disjoint. All such triangulations give small resolutions of the same contraction X_F , so a decision in one of them shows that X_F has k nodes. Toric data decide the four-node face of X_9 in this way. Every triangulation containing a face was examined (all of them when there are at most 40, otherwise a sample, as in the census). Of the 372 faces with $k \geq 3$,

toric data decide 335: all with $k = 5$, 212 of the 221 with $k = 4$ and 121 of the 149 with $k = 3$. In the others a toric divisor is contracted, so curves of zero area need not be toric: in the remaining 9 faces with $k = 4$ the light states are toric, pairwise disjoint $(-1, -1)$ -curves, but further curves of zero area are not excluded by toric data, and in the remaining 28 faces with $k = 3$ at least one light class (exactly one in 24 of them) has no toric representative in any triangulation containing the face. Of the 5233 flop walls, toric data decide 3757; in the other 1476 a toric divisor is contracted and the light class has no toric representative in the triangulations examined, so these are flops of non-toric curves. In all 5605 faces and all triangulations examined, no two zero-area toric curves meet, and every zero-area toric curve has normal degrees $(-1, -1)$ or $(-2, -2)$. Faces of the Kähler cone of the hypersurface that are not faces of a toric Kähler cone are not seen.

7.5 Other effects at finite Planck mass

Membrane instantons on three-cycles that do not vanish, and fivebrane instantons, correct the metric of the hypermultiplet manifold smoothly and preserve the gauged isometries, so they are covered by Hypothesis 3.2: the moment map of the corrected metric has no constants of integration and still vanishes at the root, since the Coulomb branch consists of supersymmetric vacua. Membrane instantons on the vanishing three-spheres are expected, as in [6, 7], not to be an additional correction but the description of the same A_3 point in terms of the neutral fields alone. The gauged torus is exact and its weights are quantised, and the charge lattice is saturated, so no discrete gauge symmetry survives at a generic Higgs vacuum.

The argument of Section 5 is five-dimensional and does not use a compactification on a circle. At finite radius the four-node face corresponds to the stratum of the four-dimensional Coulomb branch found in [4, Sects. 5.1–5.2], on which the same four hypermultiplets become massless; Section 6 treats type IIA string theory on $\widehat{X}_9$ there directly, where these states carry no Kaluza–Klein charge.

7.6 Open problems

We list the questions left open.

1. Is the metric completion at the E_2 point the Higgs branch of the interacting E_2 theory coupled to supergravity? No Lagrangian is available there; the rigid computations at the del Pezzo facet (Section 7.1) and, at another root, at the $SU(2)$ facet (Section 7.2) are consistent with defining the Higgs branch as this completion.
2. What is the Higgs branch at finite Planck mass at the $SU(2)$ facet? There the completion of the regular Higgs branch is again $N \times \mathbb{C}^2/\mathbb{Z}_4$ (conditionally, Section 7.2), but Theorem 4.5 does not apply and only the rigid model has been computed.
3. Is the metric in the uniformising chart of Definition 4.11 twice differentiable, or smooth, at the vertex, so that the transverse slice is a Riemannian orbifold there and R_0 its curvature?

4. Is there a complex-analytic version of the local model, on the twistor space or the Swann bundle? For negative scalar curvature this needs a slice theorem for pseudo-hyperkähler cones, which the known local models [17] do not provide.
5. What are the local structure of the quotient and the tangent cones of its length metric at points whose stabiliser is a proper subgroup of G , in particular along the locally Kähler and totally real pieces of [12, Sect. 5]?

A Intersection and Gopakumar–Vafa data

The intersection numbers, Gopakumar–Vafa invariants and invariant-theory computations used in this paper were carried out in SageMath from the toric data of [4]; the Gopakumar–Vafa invariants were computed from the mirror periods up to total degree eight in the dual basis. The scripts are available from the author.

Toric data. The resolution $\widehat{X}_9$ is an anticanonical hypersurface in the toric fourfold built on the polytope with vertices

$$\begin{aligned} v_0 &= (1, 0, 0, 0), \quad v_1 = (0, 1, 0, 0), \quad v_2 = (-1, -1, 0, 0), \quad v_3 = (0, 0, 1, 0), \quad v_4 = (0, 0, 0, 1), \\ v_5 &= (-4, -2, -1, 0), \quad v_6 = (-4, -2, 0, -1), \quad v_7 = (3, 1, 1, 1), \quad v_8 = (2, 1, 1, 1), \end{aligned}$$

with the triangulation of [4, Sect. 3.1]; the exceptional surface E is the divisor of the point $(-2, -1, 0, 0)$ interior to the pentagon with vertices v_3, v_4, v_5, v_6, v_8 .

All data below refer to the integral basis $H_1, \dots, H_6$ of Section 2.2 and to $J = y_2 H_2 + y_3 H_3 + y_4 H_4$. The triple intersection numbers κ_{IKM} of $\widehat{X}_9$ are those of [4, eq. (B.2)], transformed to this basis by [4, eq. (C.1)].

Cubic form and second Chern class on the face. In lexicographic order of triples from (H_2, H_3, H_4) ,

$$\kappa = (0, 2, 3, 6, 11, 17, 14, 27, 49, 83), \quad c_2 \cdot (H_2, H_3, H_4) = (36, 80, 146), \quad (\text{A.1})$$

so that

$$\frac{1}{6} J^3 = y_2^2 y_3 + 3 y_2 y_3^2 + \frac{7}{3} y_3^3 + \frac{3}{2} y_2^2 y_4 + 11 y_2 y_3 y_4 + \frac{27}{2} y_3^2 y_4 + \frac{17}{2} y_2 y_4^2 + \frac{49}{2} y_3 y_4^2 + \frac{83}{6} y_4^3. \quad (\text{A.2})$$

These agree entry by entry with the cubic form and second Chern class of the smooth side $X_{9,t}$ [4, eq. (5.26)], as they must by Proposition 2.2.

The kinetic determinant. For the 6×6 matrix $L(J)_{IK} = \kappa_{IKM} J^M$ on all of $H^2(\widehat{X}_9)$,

$$\begin{aligned} \det L(J) &= -y_2(2y_3 + 3y_4)(y_2 + 2y_4)(22y_2^2 y_3 + 96y_2 y_3^2 + 90y_3^3 + 33y_2^2 y_4 + 272y_2 y_3 y_4 \\ &\quad + 321y_3^2 y_4 + 165y_2 y_4^2 + 297y_3 y_4^2). \end{aligned} \quad (\text{A.3})$$

The last factor has positive coefficients, so $\det L < 0$ and $\det \mathcal{G} = -\frac{1}{2} \det L > 0$ on the open face. (The 3×3 block on (H_2, H_3, H_4) alone, which is all that (A.1) determines, has

a different determinant; the formula above needs the full matrix.) On the closed face the factor y_2 vanishes exactly on the del Pezzo facet $y_2 = 0$, where E collapses and its coupling degenerates. The factors $2y_3 + 3y_4$ and $y_2 + 2y_4$ are positive on $\overline{F}$ except on the edges spanned by H_2 and H_3 respectively; in particular $\det \mathcal{G} > 0$ on the relative interiors of the facets $y_3 = 0$ and $y_4 = 0$.

Massless classes on the face. The genus-zero Gopakumar–Vafa invariants of $\widehat{X}_9$ vanish on every class of zero area on F except

$$\beta_5 = C_1, \quad \beta_1 = C_2, \quad \beta_1 + \beta_5 + \beta_6 = \iota_* e_1, \quad \beta_6 = \iota_* e_2, \quad (\text{A.4})$$

each with invariant 1, among all classes of total degree at most eight in the dual basis. Proposition 2.1(1), together with the vanishing of multiple-cover and higher-genus contributions of isolated $(-1, -1)$ -curves [24, 25], gives the same conclusion in all degrees and genera.

Charge conventions and the flat quotient. Order the four light hypermultiplets as (C_1, C_2, e_1, e_2) and write $(z_i, \tilde{z}_i)$ for their scalars in one complex structure, z_i of charge q_i and $\tilde{z}_i$ of charge $-q_i$. On the three vectors (H_5, H_1, H_6) that act, the charge matrix is

$$Q = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad (\text{A.5})$$

with Smith invariants $(1, 1, 1)$ and unique relation $a = (1, 1, -1, 1)$, the relation (2.5). The complex moment maps are $\mu_{\mathbb{C}} = Q(z_1 \tilde{z}_1, \dots, z_4 \tilde{z}_4)^T$; they are invariant, so the invariants of the quotient ring are the quotient of the invariant ring. Elimination gives

$$(\mathbb{C}[z, \tilde{z}]/(\mu_{\mathbb{C}}))^{T^3} \cong \mathbb{C}[X, Y, s]/(XY + s^4), \quad X = z_1 z_2 \tilde{z}_3 z_4, \quad Y = \tilde{z}_1 \tilde{z}_2 z_3 \tilde{z}_4, \quad s = -z_3 \tilde{z}_3, \quad (\text{A.6})$$

the generators of Lemma 5.1 for this a , an A_3 surface singularity (equivalently $XY = s^4$ after $X \mapsto -X$), normal and singular only at the origin, with Hilbert series $(1 - t^8)/((1 - t^2)(1 - t^4)^2)$. Quotienting first by the third row of Q reproduces the reduced rank-one del Pezzo ring of [4] and then its rank-two flavour constraint, so the two presentations agree.

The leading correction. Proposition 5.5 has a complete proof in Section 5; the following computation is an independent check of Proposition 4.12 and of the three-dimensional image. It evaluates the quotient metric in the uniformising chart with exact truncated Taylor expansions to total degree three, including the correction ψ of Lemma 4.10, reads off the curvature of its second-order Taylor polynomial, and confirms that the terms involving ψ cancel and that the result is the restriction of the curvature to $\ell(\mathbb{H})$; the image of the T^3 -invariant tensors of hyperkähler type under $R_{\text{hk}} \mapsto R_0$ has dimension three. It was validated on two exact models. The first is $\text{TN}_\lambda \times \text{TN}_\lambda$ with the diagonal circle, where TN_λ is the Taub–NUT space, the Gibbons–Hawking space [49] with potential $\lambda + 1/(2|x|)$ and moment map x . The zero level is $x_2 = -x_1$, and the quotient is the Gibbons–Hawking space with potential $2\lambda + 1/|x|$,

which after $x = 2y$ and a rescaling of the fibre coordinate is four times $\text{TN}_{2\lambda}/\mathbb{Z}_2$, that is $\text{TN}_{\lambda/2}/\mathbb{Z}_2$ in coordinates in which the flat part of the metric is standard. Since the curvature of the model is determined only up to a rotation of the chart, the computed leading curvature is compared through its invariants: it has the invariants $(|R|^2, \text{tr } \mathcal{R}^3)$, $\mathcal{R}$ the curvature operator on two-forms, of the curvature of $\text{TN}_{\lambda/2}$ at its nut. The second uses the charges of X_9 themselves: the hyperkähler metric on $\mathbb{H}^4$ with potential matrix $\text{diag}(1/(2|x_i|)) + C$ in the form of [61], C a constant symmetric matrix (for $C = 0$ these are the toric hyperkähler metrics of [62]; for $C \neq 0$ see [63, 64]; when C is not positive semidefinite the metric is defined only near the origin, which suffices here), whose quotient by T^3 at level zero is the Gibbons–Hawking space with potential $a^\top C a + 2/|y|$, that is $\text{TN}_{a^\top C a/16}/\mathbb{Z}_4$ in coordinates in which the flat part of the metric is standard; for generic C with nonzero off-diagonal entries (several numerical choices) the computed leading curvature equals the curvature of $\text{TN}_{a^\top C a/16}$ at its nut entry by entry.

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